Computer Networks and Data Communication

Module 4

CS 212

Prof. (Dr.) SHAKTI KUNDU

Jan-May 2026

Module wise CONTENTS:

- Module 1 - Introduction: An overview of networks, the Internet, services and protocols, RFCs and standards of Networking, Data communication & its components, types of data communication and communication channels. Key components of computer network; network devices; types of computer networks; network topology; reference models (Network Architectures) – OSI & TCP/IP.
- Module 2 - Physical Layer: Introduction and functions; types of transmission media – guided and unguided; integrated services digital network (ISDN); asynchronous transfer mode (ATM).
- Module 3 - Data Link Layer: Overview; sliding window protocols - Stop-&-Wait; Go-Back-N, Selective Repeat; Error detection and correction; Hamming Code, LRC, VRC, CRC, Checksum, Round Trip Time, Latency (Delay); Medium Access Sublayer: medium access control (MAC) Addresses; Switching techniques - circuit, message & packet; Random Access Protocols - Pure ALOHA, Slotted ALOHA, CSMA / CD, CSMA / CA; Controlled Access Protocols – FDMA, TDMA, CDMA; Channelization Protocols – Reservation, Polling, Token Passing; Ethernet (802.3 IEEE standard), Wireless Links, Wireless LAN - WiFi (802.11).
- Module 4 - Network Layer: Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing Protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).
- Module 5 - Transport Layer: Introduction; transport layer protocols - UDP & TCP, TCP connection establishment; TCP connection termination; TCP congestion control. Application Layer: Introduction – features & functions; application layer protocols; Web and HTTP, Email, P2P network, its types & applications.
- Module 6 – Network Optimization, Mathematical Models, Introduction Advance Computer Networks, MANET, WSN, PCN, SDN, IoT.

- Module 4 - Network Layer: Introduction; address resolution protocol (ARP); internet protocol (IP); internet control message protocol (ICMP); internet group management protocol (IGMP); internet group management protocol (IGMP); IPv4 addressing, representation of IPv6; IPv4 vs IPv6; Conversion IPv4 to IPv6; Routing protocols - routing information protocol (RIP), open shortest path first (OSPF); border gateway protocol (BGP); Enhanced Interior Gateway Routing Protocol (EIGRP).

Network Layer – Introduction, Features & Services

- The network layer is a part of the communication process in computer networks. Its main job is to move data packets between different networks.
- It helps route these packets from the sender to the receiver across multiple paths and networks. Network-to-network connections enable the Internet to function. These connections happen at the "network layer," which sends data packets between different networks.
- In the 7-layer OSI model, the network layer is layer 3. The Internet Protocol (IP) is a key protocol used at this layer, along with other protocols for routing, testing, and encryption.

Network Layer – Introduction, Features & Services

Features of Network Layer

- The main responsibility of the Network layer is to carry the data packets from the source to the destination without changing or using them.
- If the packets are too large for delivery, they are fragmented i.e., broken down into smaller packets.
- It decides the route to be taken by the packets to travel from the source to the destination among the multiple routes available in a network (also called routing).
- The source and destination addresses are added to the data packets inside the network layer.

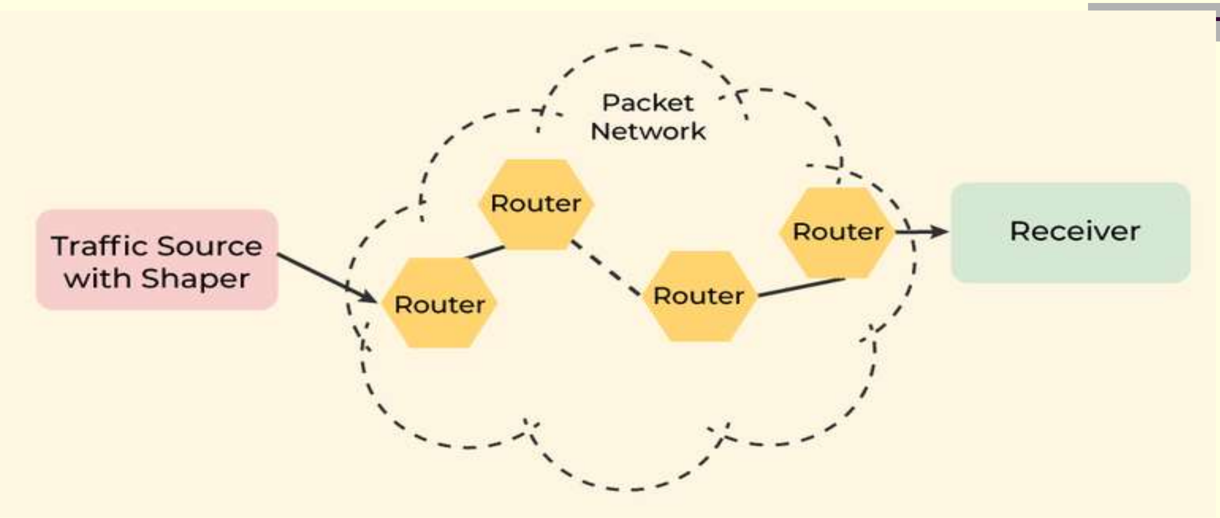
Network Layer – Introduction, Features & Services

- Services Offered by Network Layer

The services which are offered by the network layer protocol are as follows:

- Packetizing
- Routing
- Forwarding

Network Layer – Introduction, Features & Services



1. Packetizing

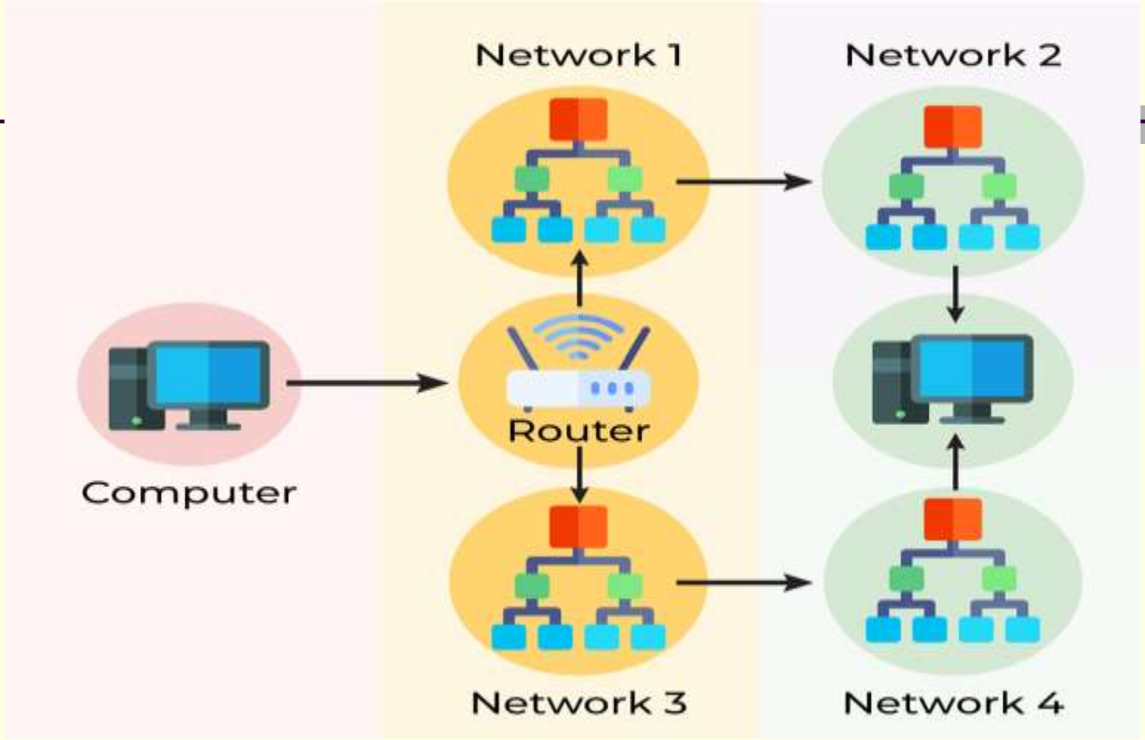
- The process of encapsulating the data received from the upper layers of the network (also called payload) in a network layer packet at the source and decapsulating the payload from the network layer packet at the destination is known as packetizing.

Network Layer – Introduction, Features & Services

1. Packetizing

- The source host adds a header that contains the source and destination address and some other relevant information required by the network layer protocol to the payload received from the upper layer protocol and delivers the packet to the data link layer.
- The destination host receives the network layer packet from its data link layer, decapsulates the packet, and delivers the payload to the corresponding upper layer protocol. The routers in the path are not allowed to change either the source or the destination address. The routers in the path are not allowed to decapsulate the packets they receive unless they need to be fragmented.

Network Layer – Introduction, Features & Services

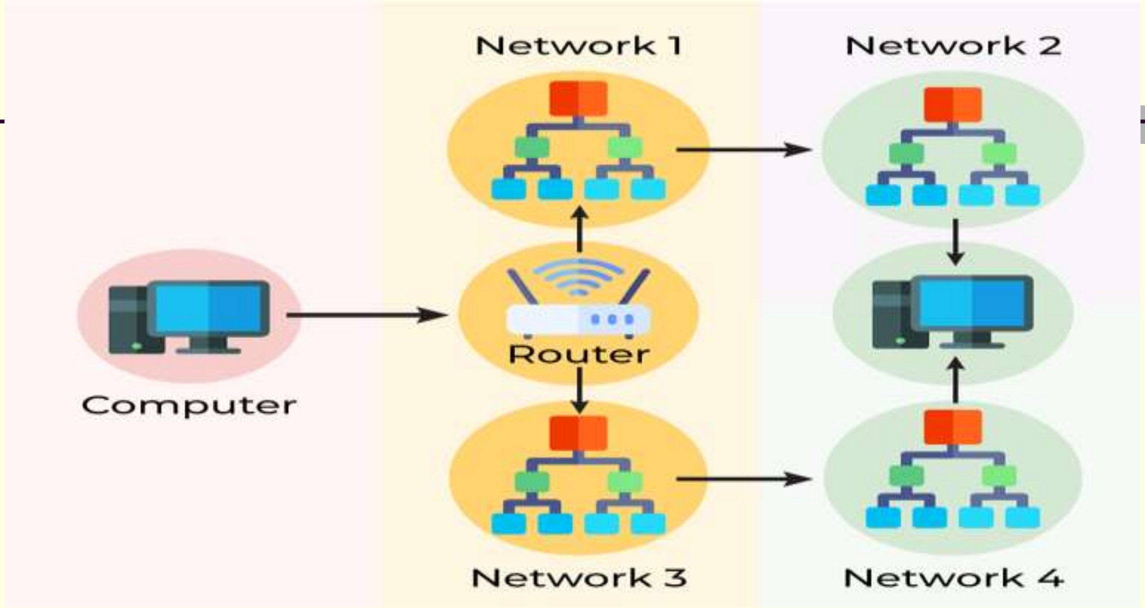


2. Routing

- **Routing is the process of moving data from one device to another device.**

  These are two other services offered by the network layer.

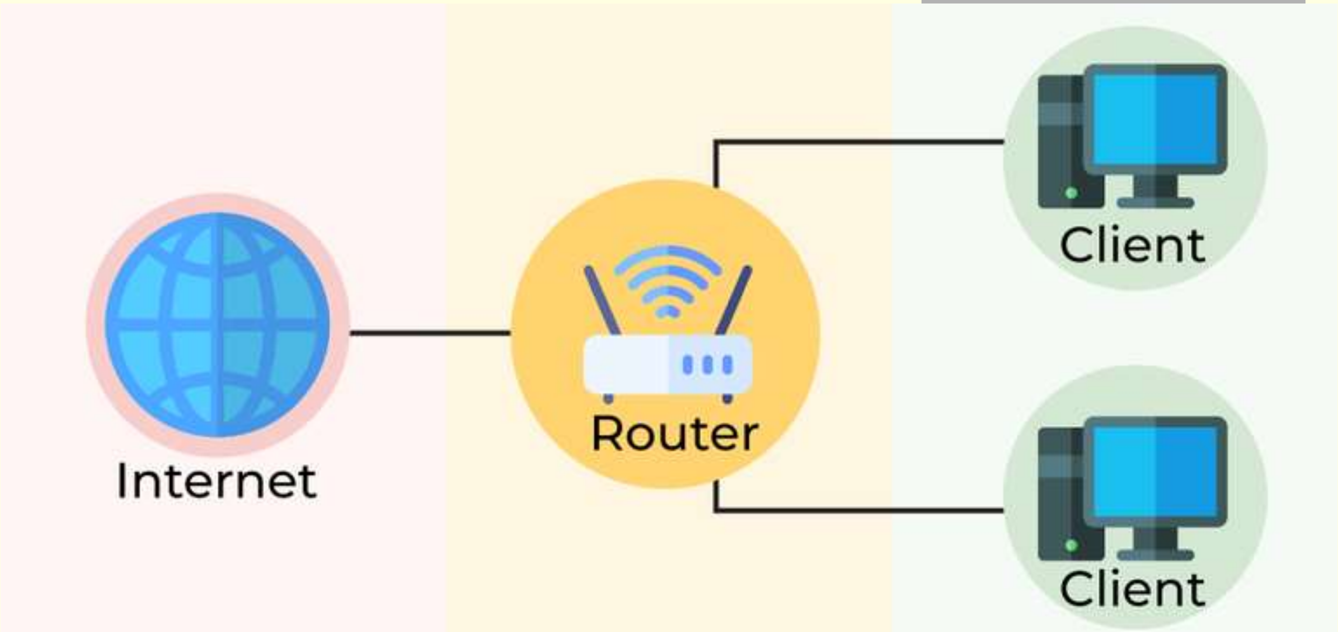
Network Layer – Introduction, Features & Services



2. Routing

- In a network, there are a number of routes available from the source to the destination. The network layer specifies some strategies which find out the best possible route. This process is referred to as **routing**. There are a number of routing protocols that are used in this process, and they should be run to help the routers coordinate with each other and help in establishing communication throughout the network.

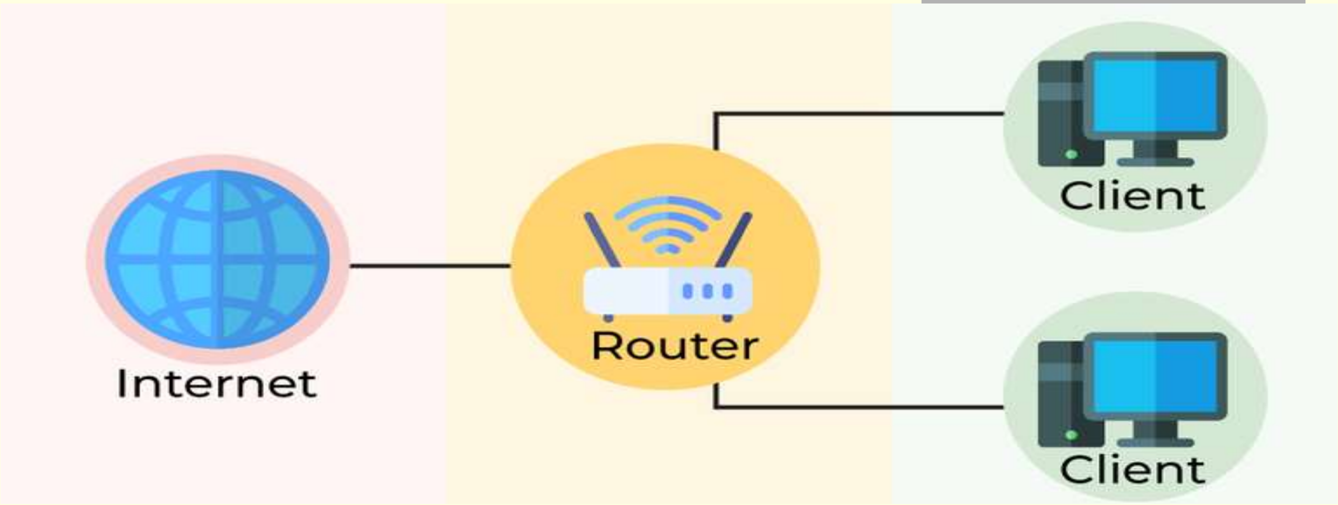
Network Layer – Introduction, Features & Services



3. Forwarding

- **Forwarding is simply defined as the action applied by each router when a packet arrives at one of its interfaces.**

Network Layer – Introduction, Features & Services



3. Forwarding

- When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network (unicast routing) or to some attached networks (in the case of multicast routing). Routers are used on the network for forwarding a packet from the local network to the remote network. So, the process of routing involves packet forwarding from an entry interface out to an exit interface.

Network Layer – Introduction, Features & Services

- Differences Between Routing and Forwarding

Routing	**Forwarding**
Routing is the process of moving data from one device to another device.	Forwarding is simply defined as the action applied by each router when a packet arrives at one of its interfaces.
Operates on the Network Layer.	Operates on the Network Layer.
Work is based on Forwarding Table.	Checks the forwarding table and work according to that.
Works on protocols like Routing Information Protocol (RIP) for Routing.	Works on protocols like UDP Encapsulating Security Payloads

Network Layer – IPv4 addressing

Outcomes

Upon the completion of this session, the learner will be able to

- ★ Understand IPv4 address in detail.
- ★ Notation of IPv4 address.
- ★ Know the Valid and Invalid IP address.

IP Address

- ★ An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a device (for example, a computer or a router) to the Internet.
- ★ An IPv4 address is 32 bits long.
- ★ Two devices on the Internet can never have the same address at the same time.
- ★ The address space of IPv4 is 2^{32} or 4,294,967,296 (more than 4 billion).

Network Layer – IPv4 addressing

Notations

- There are two prevalent notations to show an IPv4 address: binary notation and dotted decimal notation.
- Binary Notation: 01110101 10010101 00011101 00000010
- Dotted-Decimal Notation: 117.149.29.2
- Notation of IPv4 address: A.B.C.D (Only 4 octets)
- $0 \leq A,B,C,D \leq 255$
- 0.0.0.0 to 255.255.255.255

Valid IP Addresses

10.10.56.80

240.230.220.89

1.2.3.4

99.88.67.89

100.200.89.90

Invalid IP Address

56.89.1.2.5

10.065.34.56

200.28.256.8

Network Layer – IPv4 addressing

Activity Time!

Spot the error, if any, in the following IPv4 addresses.

Question	Answer
111.56.045.78	There must be no leading zero (045)
221.34.7.8.20	4 octets only in IPv4 address.
75.45.301.14	Range of each octet is between 0 and 255.
11100010.23.14.67	A mixture of binary and dotted-decimal notation is not allowed

- Formula - Conversion of dotted decimal to binary and binary to dotted decimal

Conversion (D-B and B-D)

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

Network Layer – IPv4 addressing

Example 1

Convert the IPv4 address from binary to dotted-decimal notation.
10000001 00001011 01001011 11101111

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	1

128 + 1 = 129

Example 1

Convert the IPv4 address from binary to dotted-decimal notation.
10000001 00001011 01001011 11101111

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	1

8 + 2 + 1= 11

Network Layer – IPv4 addressing

EXAMPLE 1

Convert the IPv4 address from binary to dotted-decimal notation.
10000001 00001011 01001011 11101111

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	1	0	0	1	0	1	1

$\boxed{64 + 8 + 2 + 1 = 75}$

EXAMPLE 1

Convert the IPv4 address from binary to dotted-decimal notation.
10000001 00001011 01001011 11101111

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	1	1	0	1	1	1	1

$\boxed{128 + 64 + 32 + 8 + 4 + 2 + 1 = 239}$

Network Layer – IPv4 addressing

Example 1

Convert the IPv4 address from binary to dotted-decimal notation.
10000001 00001011 01001011 11101111

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

129.11.75.239

Network Layer – IPv4 addressing

Example 2

Convert the IPv4 address from dotted-decimal to binary notation
145.14.6.8

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
1	0	0	1	0	0	0	1

1001 0001

Example 2

Convert the IPv4 address from dotted-decimal to binary notation
145.14.6.8

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	0

0000 1110

Network Layer – IPv4 addressing

EXAMPLE 2

Convert the IPv4 address from dotted-decimal to binary notation
145.14.6.8

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	0	1	1	0

0000 0110

EXAMPLE 2

Convert the IPv4 address from dotted-decimal to binary notation
145.14.6.8

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	1	0	0	0

0000 1000

Network Layer – IPv4 addressing

Example 2

Convert the IPv4 address from dotted-decimal to binary notation

145.14.6.8

SOLUTION

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	0	0	0	1	0	0	0

10010001 00001110 00000110 00001000

Network Layer – IPv4 addressing

Homework!

1. Change the following IP address from dotted-decimal notation to binary notation: 208.34.54.12

2. Change the following IP address from binary notation to dotted-decimal notation: 11101111 11110111 11000111 00011101

Classful Addressing – Subnet Mask 1 example

SUBNETTING – 5 STEPS

1. Identify the class of the IP address and note the Default Subnet Mask.
2. Convert the Default Subnet Mask into Binary.
3. Note the number of hosts required per subnet and find the Subnet Generator (SG) and octet position.
4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (subnets) in the appropriate octet position.

Classful Addressing – Subnet Mask 1 example

QUESTION

Subnet the IP address 216.21.5.0 into 30 hosts in each subnet.

SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 30 (11110) – 5 bits SG: 32 Octet Position: 4

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0
4. New subnet mask: 255.255.255.224 or /27
5. Network Ranges (Subnets)

   216.21.5.0 – 216.21.5.31
   216.21.5.32 – 216.21.5.63
   216.21.5.64 – 216.21.5.95
   216.21.5.96 – 216.21.5.127
   216.21.5.128 – 216.21.5.159
   and so on....

- First IP address of the subnet represents Network address whereas the last IP address of the subnet represents the Broadcast address.
- Here, 216.21.5.31 is the broadcast address of first subnet which is usually not assigned to the Host.
- Within the network, SWITCH is sufficient to establish a communication.
- On the other hand, ROUTER is needed to establish a communication in different network.

Classful Addressing – Subnet Mask 2 example

QUESTION

Subnet the IP address 196.10.20.0 into 52 hosts in each subnet.

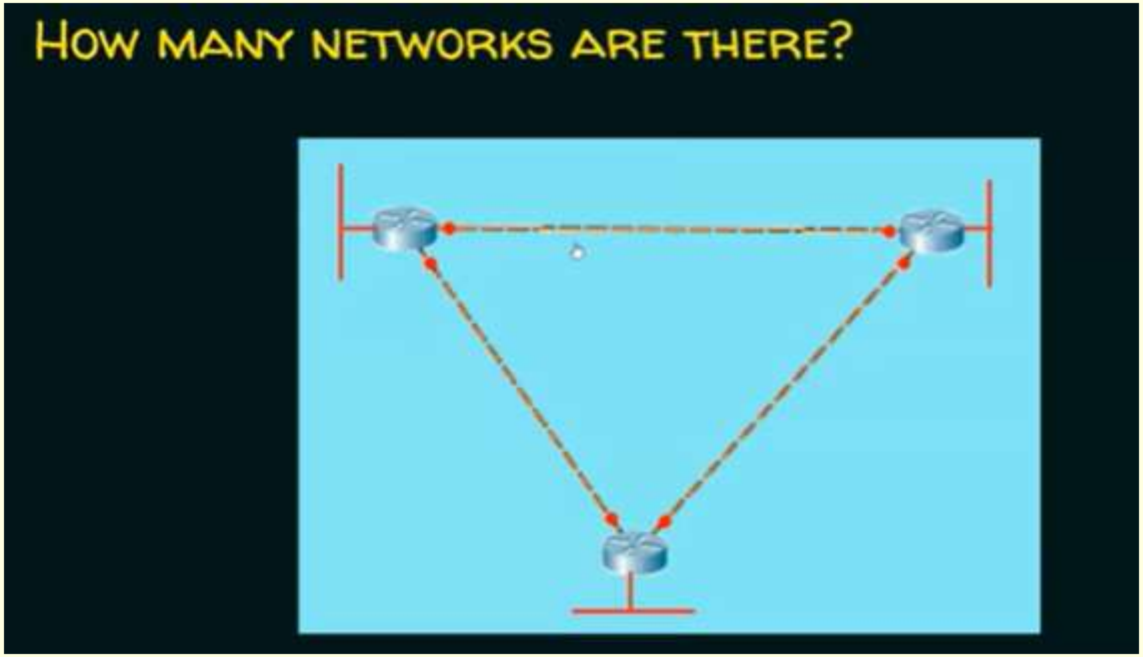
SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 52 (110100) – 6 bits SG: 64 Octet Position: 4

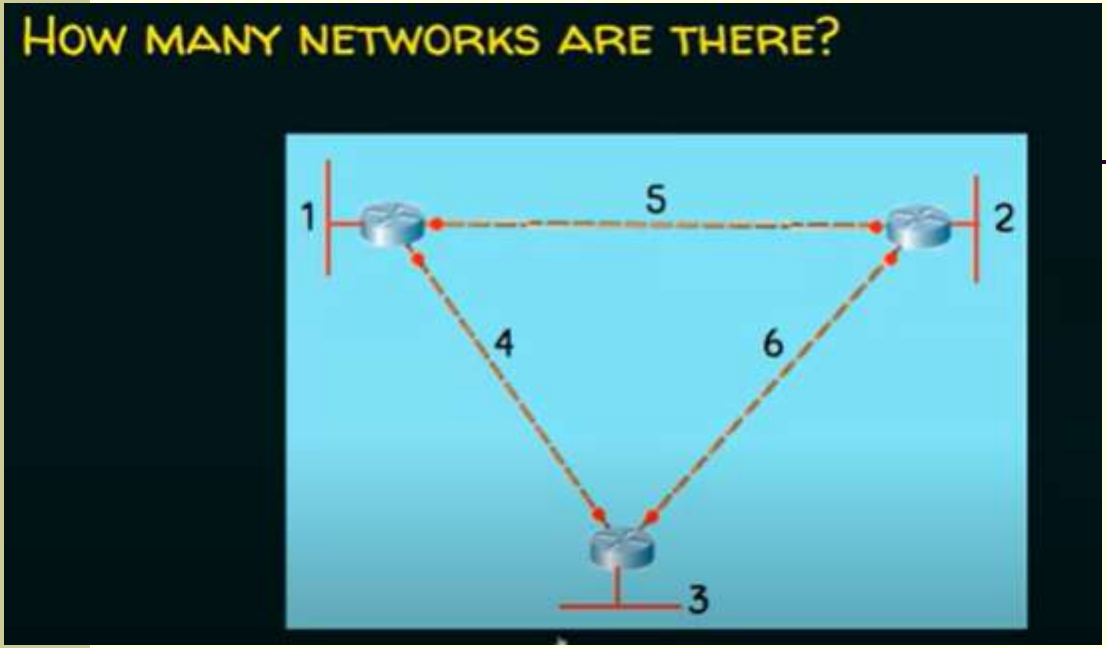
   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0
4. New subnet mask: 255.255.255.192 or /26
5. Network Ranges (Subnets)

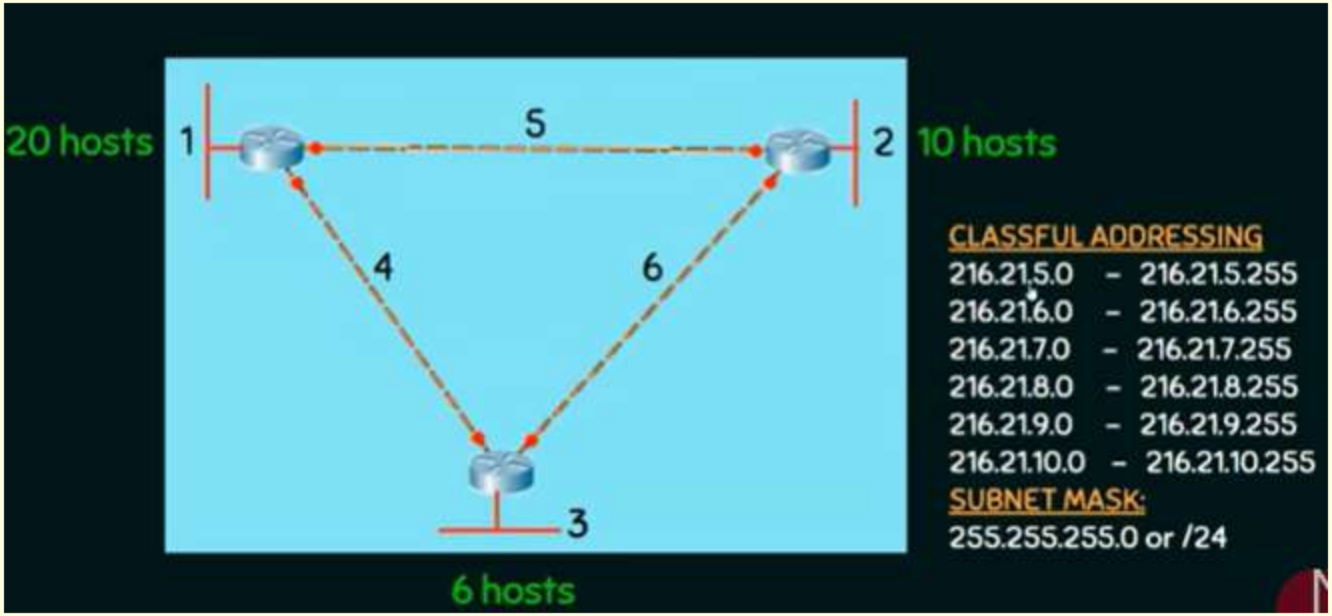
   196.10.20.0 – 196.10.20.63
   196.10.20.64 – 196.10.20.127
   196.10.20.128 – 196.10.20.191
   196.10.20.192 – 196.10.20.255

Fixed Lenth & Variable Length Subnet Masking

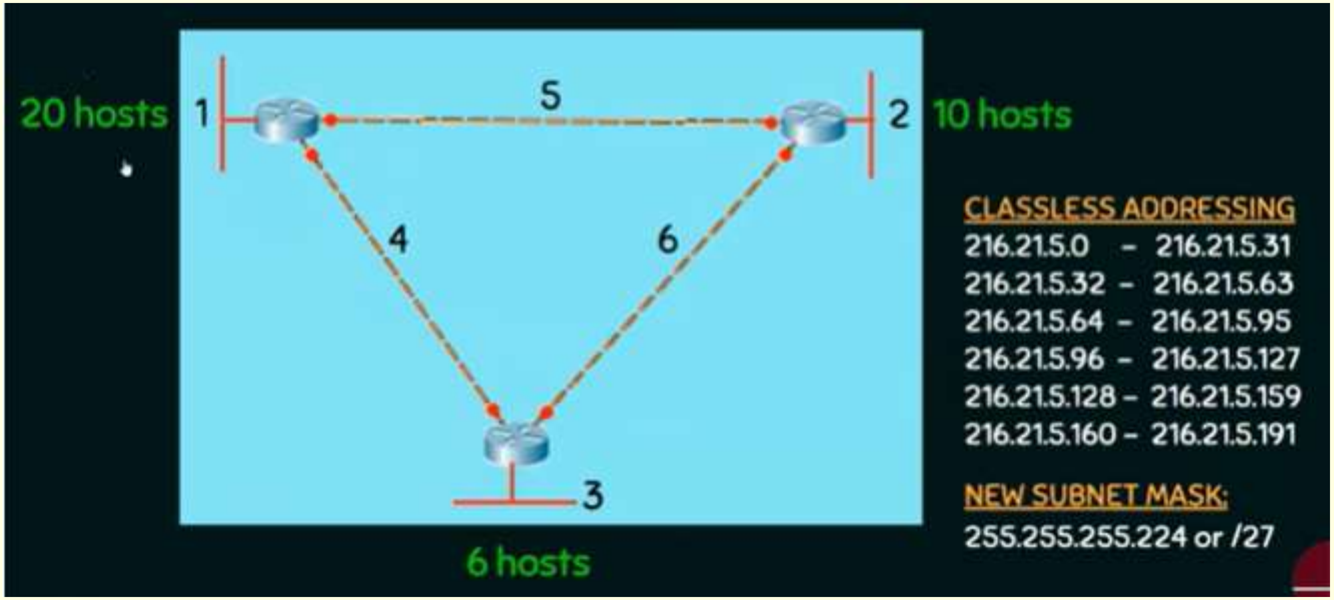


Fixed Lenth Subnet Masking

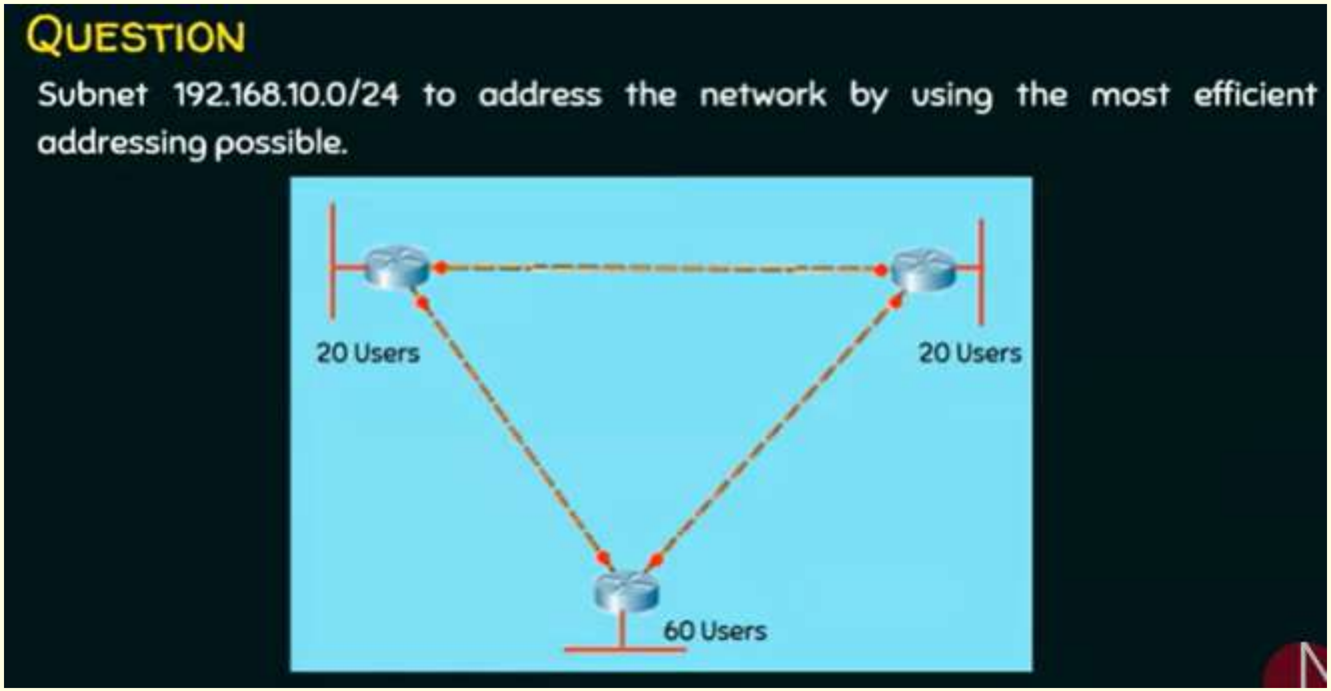




Fixed Lenth Subnet Masking – another example



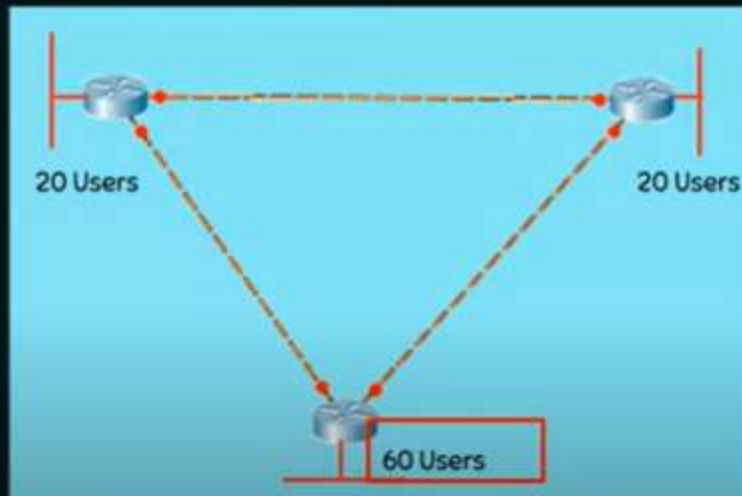
Variable Length Subnet Masking



Variable Length Subnet Masking

Key Idea

Start with the largest subnet first



Subnetting – 5 Steps

1. Identify the class of the IP address and note the Default Subnet Mask.
2. Convert the Default Subnet Mask into Binary.
3. Note the number of hosts required per subnet and find the Subnet Generator (SG) and octet position.
4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (subnets) in the appropriate octet position.

Variable Length Subnet Masking

Solution

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 60 (111100) – 6 bits SG: 64 Octet Position: 4

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 0 0 0 0 0 0
4. New subnet mask: 255.255.255.192 or /26 (Only for the biggest network)
5. Network Ranges (Subnets)

   192.168.10.0 – 192.168.10.63 /26 (Handover this to 60 Users Network)
   192.168.10.64

Solution

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 20 (10100) – 5 bits SG: 32 Octet Position: 4

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0
4. New subnet mask: 255.255.255.224 or /27
5. Network Ranges (Subnets)

   192.168.10.0 – 192.168.10.63 /26 (Handover this to 60 Users Network)
   192.168.10.64 – 192.168.10.95 /27 (Handover this to 20 Users Network)

Variable Length Subnet Masking

SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 20 (10100) – 5 bits SG: 32 Octet Position: 4

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0
4. New subnet mask: 255.255.255.224 or /27
5. Network Ranges (Subnets)

   192.168.10.0 – 192.168.10.63 /26 (Handover this to 60 Users Network)
   192.168.10.64 – 192.168.10.95 /27 (Handover this to 20 Users Network)
   192.168.10.96 – 192.168.10.127 /27 (Handover this to another 20 Users Network)

SOLUTION

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: 2 (10) – 2 bits SG: 4 Octet Position: 4

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0
4. New subnet mask: 255.255.255.252 or /30
5. Network Ranges (Subnets)

   192.168.10.0 – 192.168.10.63 /26 (Handover this to 60 Users Network)
   192.168.10.64 – 192.168.10.95 /27 (Handover this to 20 Users Network)
   192.168.10.96 – 192.168.10.127 /27 (Handover this to another 20 Users Network)
   192.168.10.128–192.168.10.131 / 30 (Handover this to Crossover Link)

Variable Length Subnet Masking

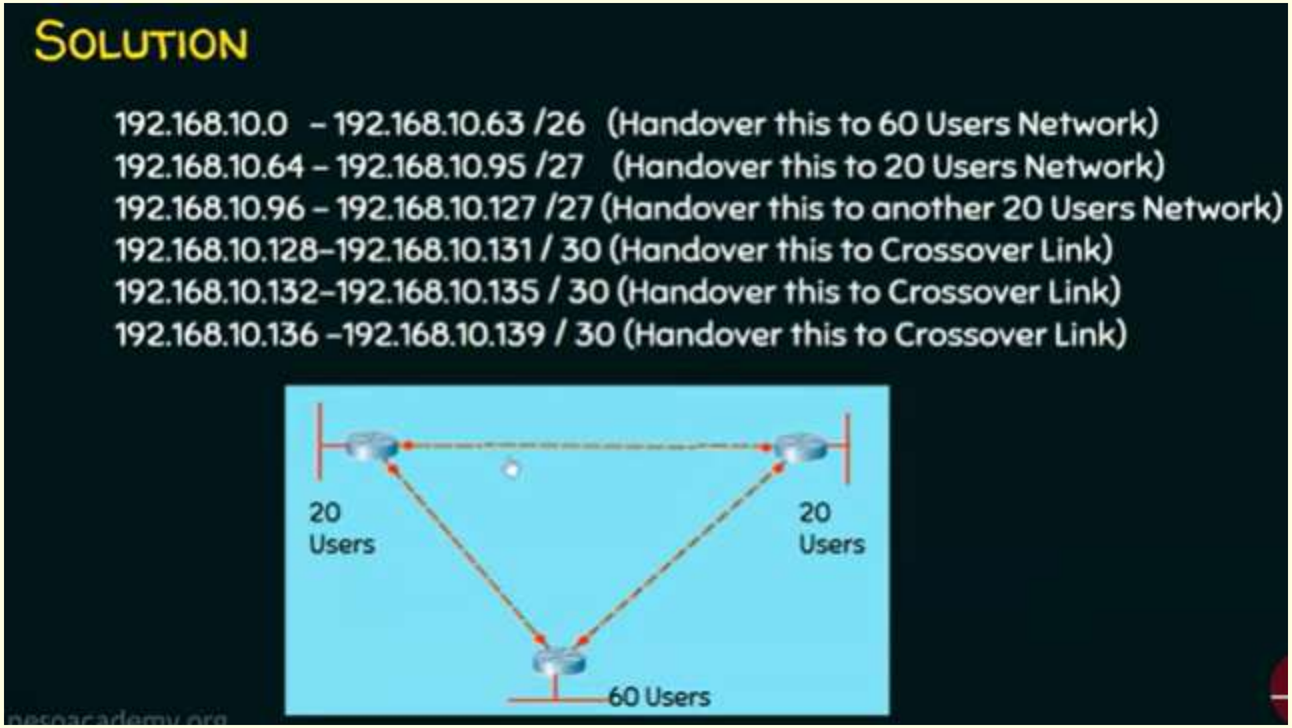
Solution

1. Class C – Default Subnet Mask: 255.255.255.0
2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
3. No. of hosts/subnet: $\boxed{\text{2 (10) – 2 bits}}$ SG: $\boxed{4}$ Octet Position: $\boxed{4}$

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0
4. New subnet mask: 255.255.255.252 or /30
5. Network Ranges (Subnets)

   192.168.10.0 – 192.168.10.63 /26 (Handover this to 60 Users Network)
   192.168.10.64 – 192.168.10.95 /27 (Handover this to 20 Users Network)
   192.168.10.96 – 192.168.10.127 /27 (Handover this to another 20 Users Network)
   192.168.10.128–192.168.10.131 / 30 (Handover this to Crossover Link)
   192.168.10.132–192.168.10.135 / 30 (Handover this to Crossover Link)
   192.168.10.136 –192.168.10.139 / 30 (Handover this to Crossover Link)

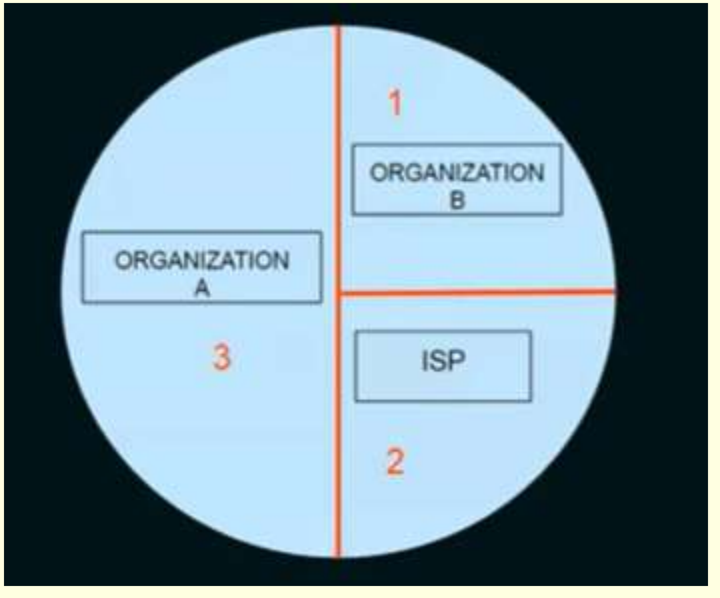
Variable Length Subnet Masking



Variable Length Subnet Masking – Example 2

QUESTION

An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?



Variable Length Subnet Masking

SOLUTION

1. Subnet Mask: /20

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0 . 0 0 0 0 0 0 0 0

2. No. of hosts = $2^{\text{No. of 0's}} = 2^{12}$
3. Handover quarter of the address to Organization B = $2^{12}/4 = 2^{12}/2^{2} = 2^{10}$

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0 . 0 0 0 0 0 0 0 0

   SG=4 , OP=3, Starting Address = 245.248.128.0 /22 to 245.248.131.255 /22

4. Handover another quarter of address to ISP itself.

   SG=4, OP=3, Starting Address = 245.248.132.0 / 22 to 245.248.135.255 /22

Variable Length Subnet Masking

SOLUTION

1. Subnet Mask: /20

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 0 0 0 0 . 0 0 0 0 0 0 0 0

2. No. of hosts = $2^{\text{No. of 0's}} = 2^{12}$
3. Handover quarter of the address to Organization B = $2^{12}/4 = 2^{12}/2^{2} = 2^{10}$

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0 . 0 0 0 0 0 0 0 0

   SG=4 , OP=3, Starting Address = 245.248.128.0 /22 to 245.248.131.255 /22

4. Handover another quarter of address to ISP itself.

   SG=4, OP=3, Starting Address = 245.248.132.0 / 22 to 245.248.135.255 /22

5. Handover half of the addresses to Organization A = $2^{12}/2 = 2^{11}$

   1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 0 0 0 . 0 0 0 0 0 0 0 0

   SG=8, OP=3, Starting Address = 245.248.136.0 /21 to 245.248.143.255 /21

Network Layer – IPv4

- IP stands for Internet Protocol version v4 (IPv4), is the most widely used system for identifying devices on a network. It uses a set of four numbers, separated by periods (like 192.168.0.1), to give each device a unique address. This address helps data find its way from one device to another over the internet.
- IPv4 was the primary version brought into action for production within the ARPANET in 1983. IP version four addresses are 32-bit integers which will be expressed in decimal notation. Example- 192.0.2.126 could be an IPv4 address.

Network Layer – IPv4

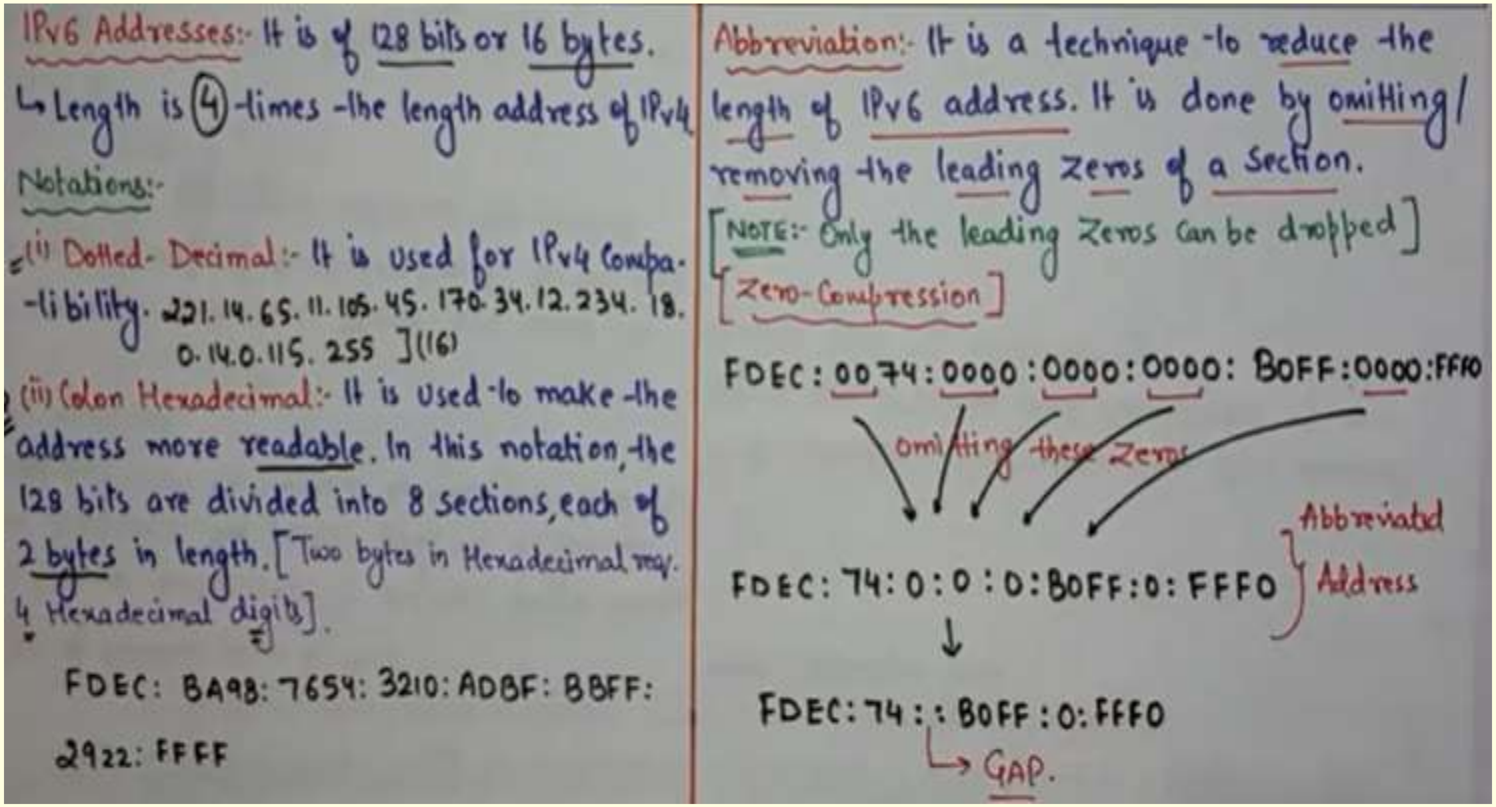
Parts of IPv4: IPv4 addresses consist of three parts:

- Network Part: The network part indicates the distinctive variety that's appointed to the network. The network part conjointly identifies the category of the network that's assigned.
- Host Part: The host part uniquely identifies the machine on your network. This part of the IPv4 address is assigned to every host. For each host on the network, the network part is the same, however, the host half must vary.
- Subnet Number: This is the nonobligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets and subnet numbers are appointed to that.

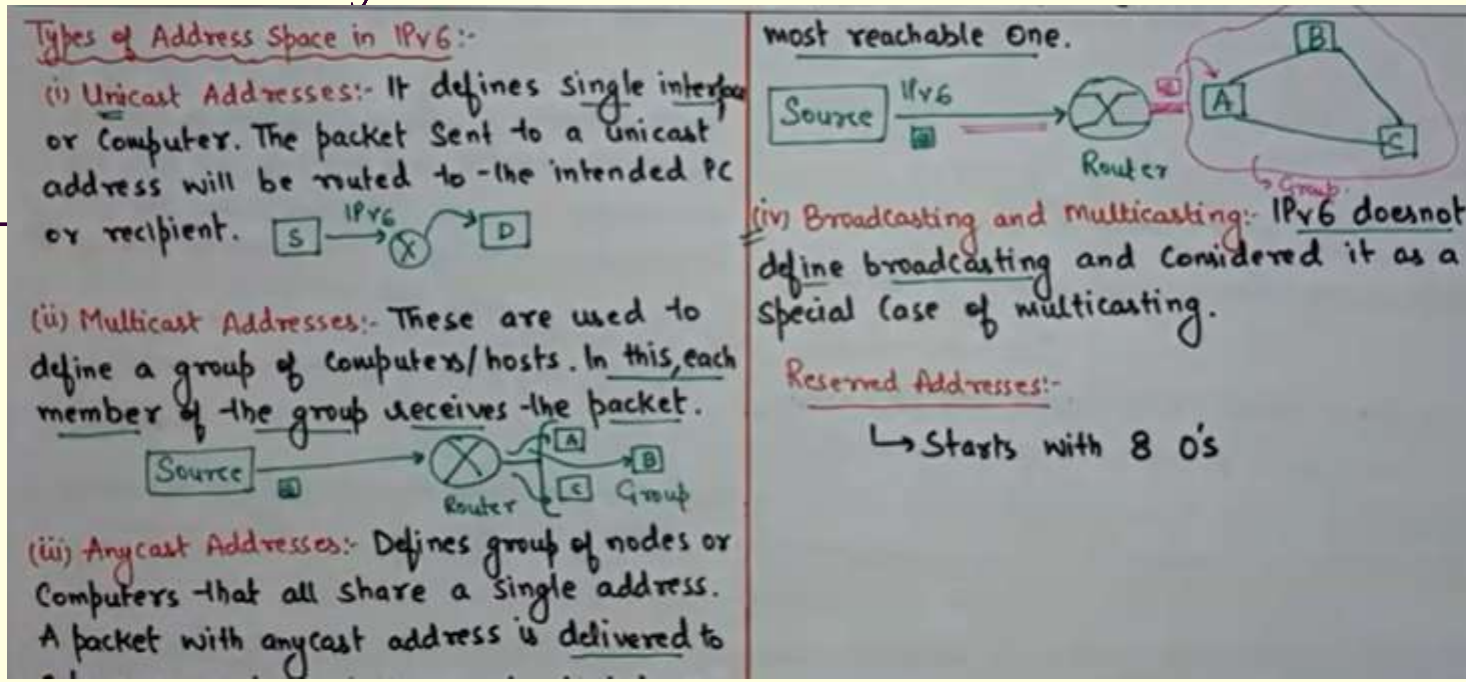
Network Layer – IPv4

- Characteristics of IPv4
  - IPv4 could be a 32-bit IP Address.
  - IPv4 could be a numeric address, and its bits are separated by a dot.
  - The number of header fields is twelve and the length of the header field is twenty.
  - It has Unicast, broadcast, and multicast-style addresses.
  - IPv4 supports VLSM (Virtual Length Subnet Mask).
  - IPv4 uses the Post Address Resolution Protocol to map to the MAC address.
  - RIP may be a routing protocol supported by the routed daemon.
  - Networks ought to be designed either manually or with DHCP.
  - Packet fragmentation permits from routers and causes host.

Network Layer – IPv6



Network Layer – IPv6



Types of Address Space in IPv6:-

(i) Unicast Addresses:- It defines single interface or Computer. The packet Sent to a unicast address will be routed to the intended PC or recipient.

S → IPv6 → X → D

(ii) Multicast Addresses:- These are used to define a group of computers/hosts. In this, each member of the group receives the packet.

Source → Router → A, B, C (Group)

(iii) Anycast Addresses:- Defines group of nodes or Computers that all share a single address. A packet with anycast address is delivered to

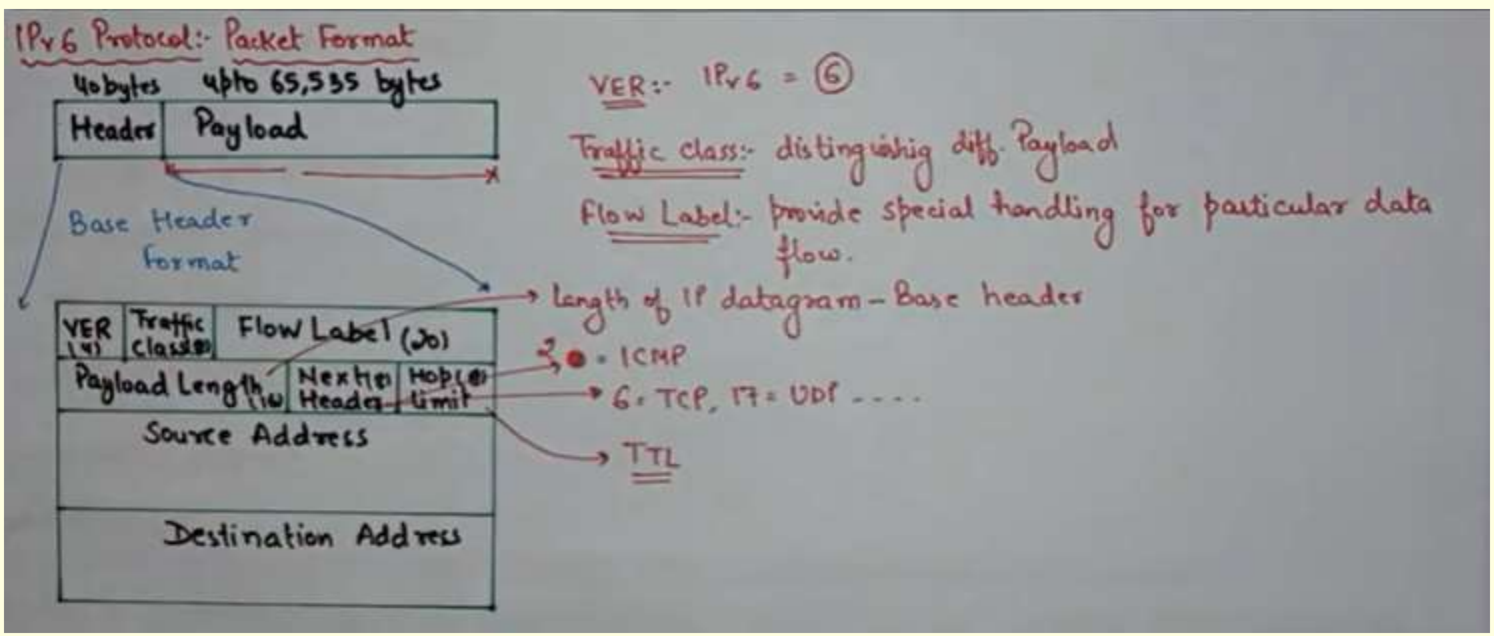
most reachable one.

Source → IPv6 → Router → A, B, C (Group)

(iv) Broadcasting and Multicasting:- IPv6 doesnot define broadcasting and Considered it as a special case of multicasting.

Reserved Addresses:-

↳ Starts with 8 0's



IPv6 Protocol:- Packet Format

40 bytes	upto 65,535 bytes
Header	Payload

Base Header Format

VER (4)	Traffic Class (8)	Flow Label (20)	
Payload Length (16)		Next Header (8)	Hop Limit (8)
Source Address			
Destination Address			

VER:- IPv6 = 6

Traffic class:- distinguishing diff. Payload

Flow Label:- provide special handling for particular data flow.

→ Length of IP datagram − Base header

= ICMP

→ 6 = TCP, 17 = UDP ....

→ TTL

Network Layer – IPv6

Changes in IPv6:-

i) Larger address space ⇒ $2^{128} - 2^{32}$ ⇒ 2^{96} time more address than IPv4.

ii) Better Header format ⇒ options are separated from Base Header.

iii) New options ⇒ Additional functionalities.

iv) Allowance for Extension ⇒ Extension of Protocol

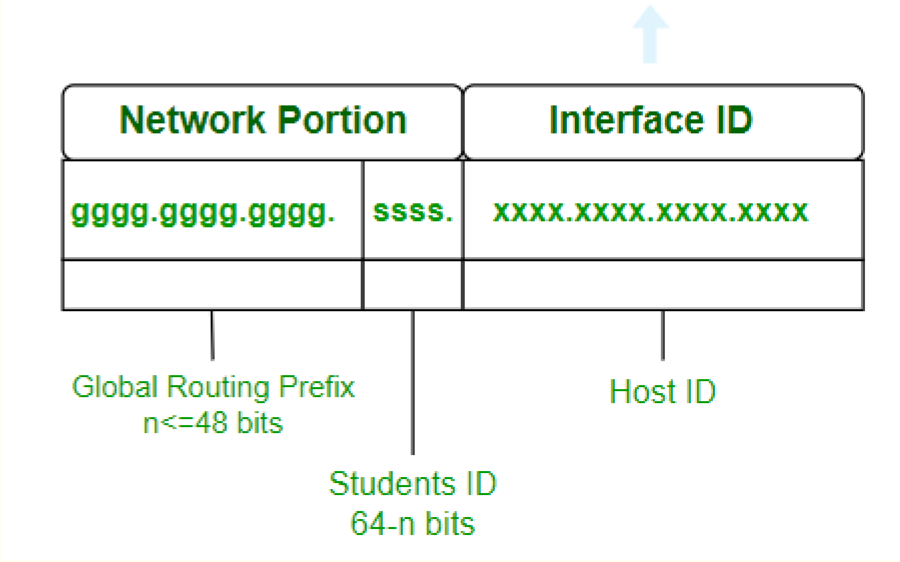
v) Support for Resource Allocation ⇒ Traffic class, Flow Label → Special Handling of the packet

vi) Support for More Security ⇒ Encryption and Authentications options.

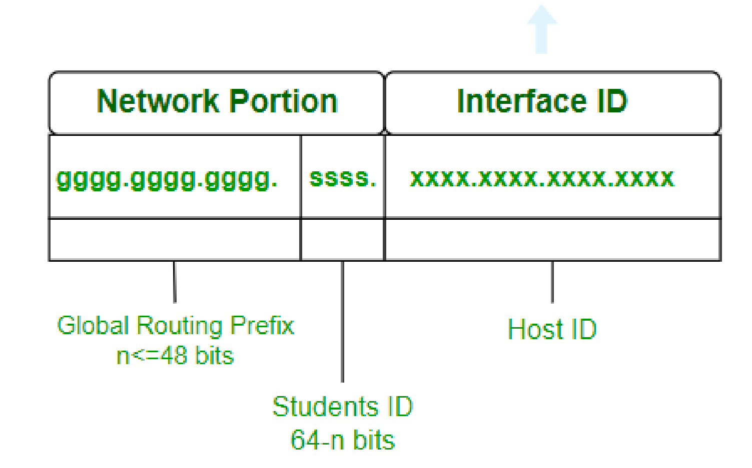
Network Layer – IPv6

- The next generation Internet Protocol (IP) address standard, known as IPv6, is meant to work in tandem with IPv4, which is still in widespread use today, and eventually replace it. To communicate with other devices, a computer, smartphone, home automation component, Internet of Things sensor, or any other Internet-connected device needs a numerical IP address. Because so many connected devices are being used, the original IP address scheme, known as IPv4, is running out of addresses.
- Representation of IPv6

An IPv6 address consists of eight groups of four hexadecimal digits separated by ' . ' and each Hex digit representing four bits so the total length of IPv6 is 128 bits. Structure given below.



Network Layer – IPv6



gggg.gggg.gggg.ssss.xxxx.xxxx.xxxx.xxxx

- The first 48 bits represent Global Routing Prefix. The next 16 bits represent the student ID and the last 64 bits represent the host ID. The first 64 bits represent the network portion and the last 64 bits represent the interface id.

- Global Routing Prefix: The Global Routing Prefix is the portion of an IPv6 address that is used to identify a specific network or subnet within the larger IPv6 internet. It is assigned by an ISP or a regional internet registry (RIR).
- Student Id: The portion of the address used within an organization to identify subnets. This usually follows the Global Routing Prefix.
- Host Id: The last part of the address, is used to identify a specific host on a network.
- Example: 3001:0da8:75a3:0000:0000:8a2e:0370:7334

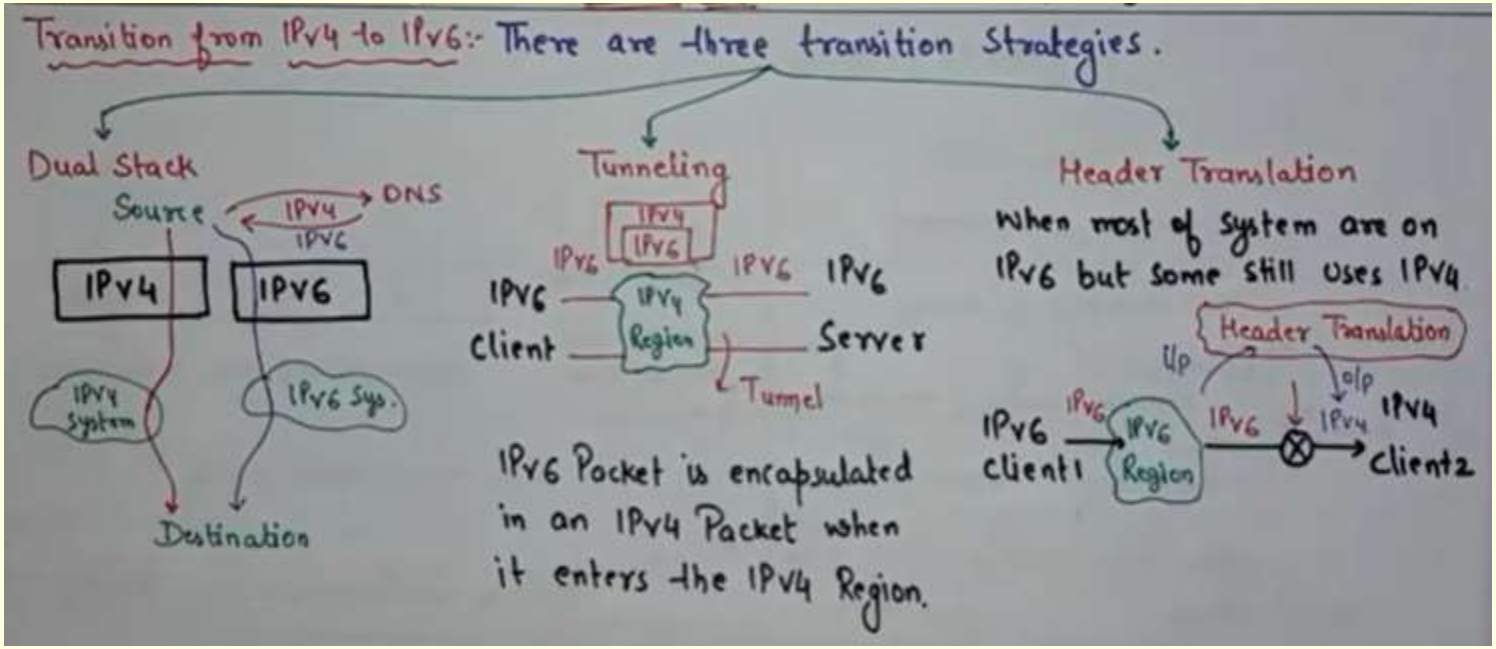
Network Layer – IPv6

- Types of IPv6 Address
- Unicast Addresses: Only one interface is specified by the unicast address. A packet moves from one host to the destination host when it is sent to a unicast address destination.
- Multicast Addresses: It represents a group of IP devices and can only be used as the destination of a datagram.
- Anycast Addresses: The multicast address and the anycast address are the same. The way the anycast address varies from other addresses is that it can deliver the same IP address to several servers or devices. Keep in mind that the hosts do not receive the IP address. Stated differently, multiple interfaces or a collection of interfaces are assigned an anycast address.

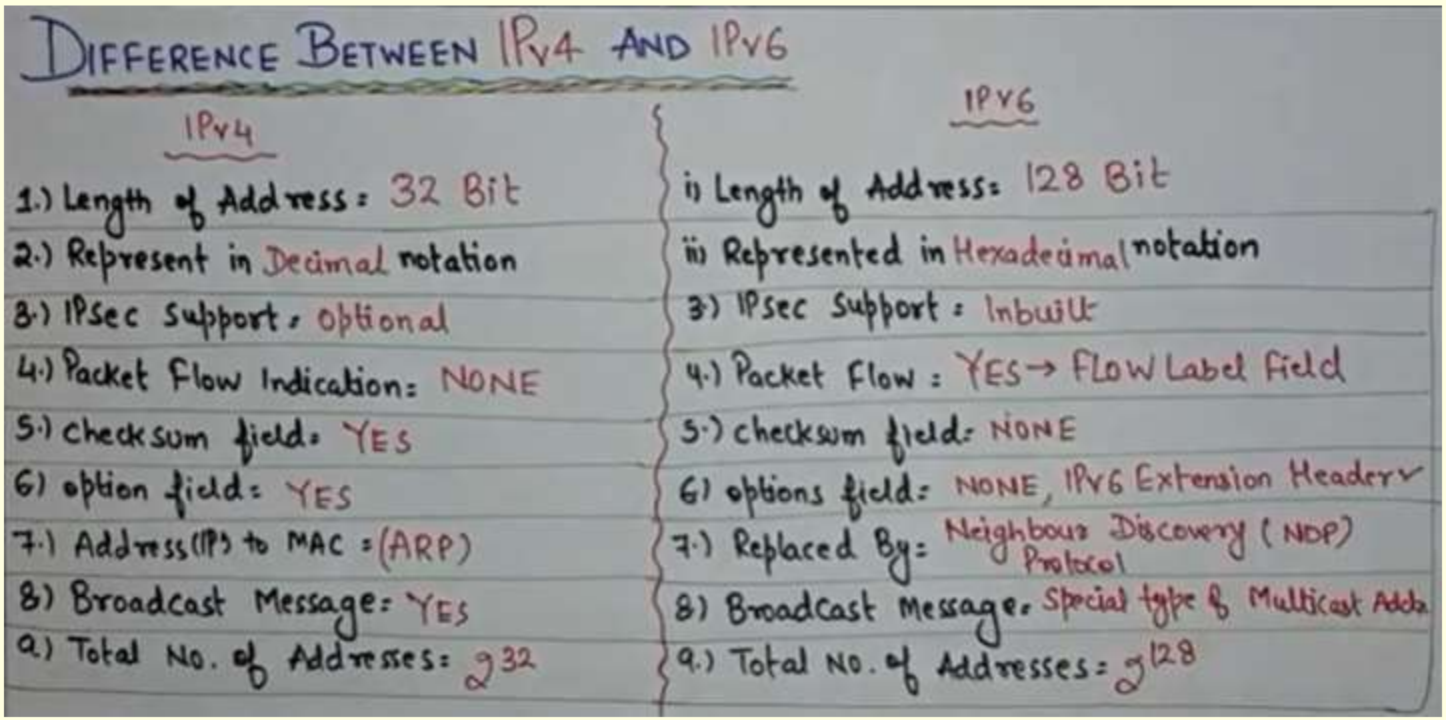
IPv6 vs IPv4

- This new IP address version is being deployed to fulfil the need for more Internet addresses. With 128-bit address space, it allows 340 undecillion unique address space.
- IPv6 support a theoretical maximum of 340, 282, 366, 920, 938, 463, 463, 374, 607, 431, 768, 211, 456. To keep it straightforward, we will never run out of IP addresses again.

Conversion of IPv4 to IPv6



Difference between IPv4 & IPv6

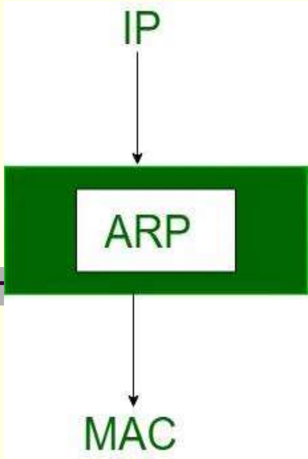


Network Layer – IPv6

- The next iteration of the IP standard is known as Internet Protocol version 6 (IPv6). Although IPv4 and IPv6 will coexist for a while, IPv6 is meant to work in tandem with IPv4 before eventually taking its place. We need to implement IPv6 in order to proceed and keep bringing new gadgets and services to the Internet.

IPv6	**IPv4**
IPv6 has a 128-bit address length	IPv4 has a 32-bit address length
It supports Auto and renumbering address configuration	It Supports Manual and DHCP address configuration
The address space of IPv6 is quite large it can produce 3.4×10^{38} address space	It can generate 4.29×10^{9} address space
Address Representation of IPv6 is in hexadecimal	Address representation of IPv4 is in decimal
In IPv6 checksum field is not available	In IPv4 checksum field is available
IPv6 has a header of 40 bytes fixed	IPv4 has a header of 20-60 bytes.
IPv6 does not support VLSM.	IPv4 supports VLSM(Variable Length subnet mask).

Network Layer – Address Resolution Protocol



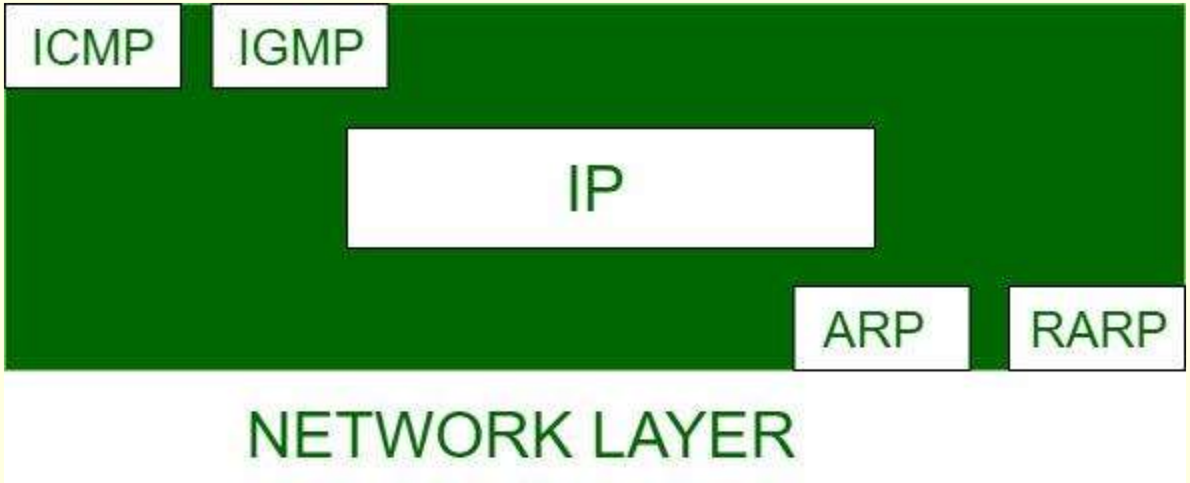
- When computer programs send or get messages, they usually use something called an IP address, which is like a virtual address. But underneath, the real talk happens using another type of address called a MAC address, which is like a device's actual home address.

- So, our goal is to find out the MAC address of where we want to talk to. That's where ARP comes in handy. **It helps by turning the IP address into the physical MAC address, so we can chat with other devices on the network.**

- Most computer programs/applications use logical addresses (IP Addresses) to send/receive messages. However, the actual communication happens over the Physical Address (MAC Address) from layer 2 of the OSI model. So, our mission is to get the destination MAC Address which helps communicate with other devices. This is where ARP comes into the picture; its functionality is to translate IP addresses into physical addresses.

Network Layer – Address Resolution Protocol

- The acronym ARP stands for Address Resolution Protocol which is one of the most important protocols of the Data link layer in the OSI model. It is responsible to find the hardware address of a host from a known IP address. There are three basic ARP terms:
  - Reverse ARP
  - Proxy ARP
  - Inverse ARP



Network Layer – Address Resolution Protocol

Reverse ARP

- Reverse Address Resolution Protocol is a protocol that is used in local area networks (LAN) by client machines for requesting IP Address (IPv4) from Router's ARP Table. Whenever a new machine comes, which requires an IP Address for its use. In that case, the machine sends a RARP broadcast packet containing MAC Address in the sender and receiver hardware field.

Proxy ARP

- Proxy Address Resolution Protocol work to enable devices that are separated into network segments connected through the router in the same IP to resolve IP Address to MAC Address. Proxy ARP is enabled so that the 'proxy router' resides with its MAC address in a local network as it is the desired router to which broadcast is addressed. In case, when the sender receives the MAC Address of the Proxy Router, it is going to send the datagram to Proxy Router, which will be sent to the destination device.

Network Layer – Address Resolution Protocol

Inverse ARP

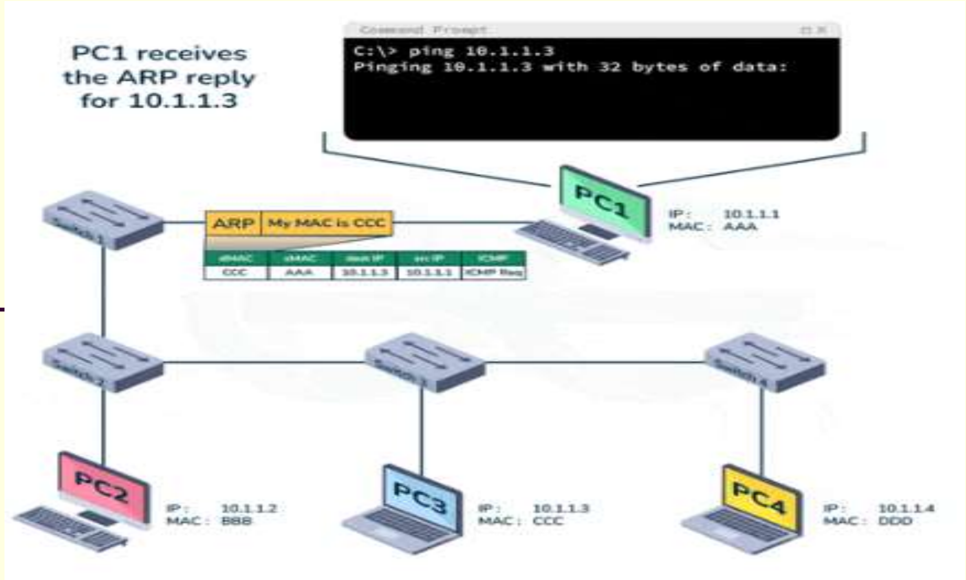
- Inverse Address Resolution Protocol uses MAC Address to find the IP Address, it can be simply illustrated as Inverse ARP is just the inverse of ARP. In ATM (Asynchronous Transfer Mode) Networks, Inverse ARP is used by default. Inverse ARP helps in finding Layer-3 Addresses from Layer-2 Addresses.

Network Layer – Address Resolution Protocol

How ARP Works?

- Imagine a device that wants to communicate with others over the internet. What does ARP do? It broadcast a packet to all the devices of the source network. The devices of the network peel the header of the data link layer from the Protocol Data Unit (PDU) called frame and transfer the packet to the network layer (layer 3 of OSI) where the network ID of the packet is validated with the destination IP's network ID of the packet and if it's equal then it responds to the source with the MAC address of the destination, else the packet reaches the gateway of the network and broadcasts packet to the devices it is connected with and validates their network ID. The above process continues till the second last network device in the path reaches the destination where it gets validated and ARP, in turn, responds with the destination MAC address.

Network Layer – Address Resolution Protocol



- ARP Cache: After resolving the MAC address, the ARP sends it to the source where it is stored in a table for future reference. The subsequent communications can use the MAC address from the table.
- ARP Cache Timeout: It indicates the time for which the MAC address in the ARP cache can reside.
- ARP request: This is nothing but broadcasting a packet over the network to validate whether we came across the destination MAC address or not.
  - The physical address of the sender.
  - The IP address of the sender.
  - The physical address of the receiver is FF:FF:FF:FF:FF: FF or 1's.
  - The IP address of the receiver.
- ARP response/reply: It is the MAC address response that the source receives from the destination which aids in further communication of the data.

Network Layer – Address Resolution Protocol

- CASE-1:
  - The sender wants to send a packet to another host on the same network.
  - Use ARP to find another host's physical address.
- CASE-2:
  - The sender is a host and wants to send a packet to another host on another network.
  - The sender looks at its routing table.
  - Find the IP address of the next hop (router) for this destination.
- CASE-3:
  - The sender is a router and received a datagram.
  - The router checks its routing table.
  - Calculate the IP of the next router.
  - Use ARP to find the next router's physical address.
- CASE-4:
  - The sender is a router that has received a datagram destined for a host in the same network.
  - Use ARP to find this host's physical address.
- Note: An ARP request is broadcast, and an ARP response is a Unicast.

Network Layer – Address Resolution Protocol

- Connect two PC, say A and B with a cross cable. Now you can see the working of ARP by typing these commands:

  1. A > arp -a

- There will be no entry at the table because they never communicated with each other.

  2. A > ping 192.168.1.2

  IP address of destination is 192.168.1.2
  Reply comes from destination but one packet is lost because of ARP processing.

Network Layer – Address Resolution Protocol

```
PC>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=0ms TTL=255
Reply from 192.168.1.2: bytes=32 time=0ms TTL=255
Reply from 192.168.1.2: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- Now, entries of the ARP table can be seen by typing the command. This is what the ARP table looks like:

```
C:\Users\[illegible]>arp -a

Interface: 192.168.1.113 --- 0x3
  Internet Address      Physical Address      Type
  192.168.1.100         [illegible]           dynamic
  192.168.1.101         [illegible]           dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  [illegible]           [illegible]           static
  224.0.0.251           [illegible]           static
  [illegible]           [illegible]           static
  [illegible]           [illegible]           static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

Network Layer – Address Resolution Protocol

Types of ARP

1. Proxy ARP

- A Layer 3 device can reply to an ARP request for a target that is on a different network than the sender by using a technique called proxy ARP. In response to the ARP, the router that has been set for Proxy ARP maps its MAC address to the target IP address, deceiving the sender into believing that the message has arrived at destination.

Because the packets have the required information, the proxy router at the backend forwards them to the correct location.

Network Layer – Address Resolution Protocol

Types of ARP

2. Gratuitous ARP

- The host's ARP request known as "gratuitous ARP" aids in locating duplicate IP addresses. This is a broadcast request for the router's IP address. All other nodes are unable to use the IP address assigned to a switch or router in the event that it sends out an ARP request to obtain its IP address and receives no ARP answers in return. However, another node uses the IP address assigned to the switch or router if it sends an ARP request for its IP address and gets an ARP response.

Network Layer – Address Resolution Protocol

Types of ARP

3. Reverse ARP

- In a local area network (LAN), the client system uses this networking protocol to ask the ARP gateway router table for its IPv4 address. The network administrator creates a table in the gateway-router that is used to correlate the IP address with the MAC address.

4. Inverse ARP

- The purpose of inverse ARP, which is the opposite of ARP, is to deduce the nodes' IP addresses from their data link layer addresses. Frame relays and ATM networks, where Layer 2 virtual circuit addressing is frequently obtained from Layer 2 signaling, are the primary applications for them. These virtual circuits can be used with the necessary Layer 3 addresses accessible.

Network Layer - ICMP

- **ICMP** is used for reporting errors and management queries.
- **It is a supporting protocol and is used by network devices like routers for sending error messages and operations information.** For example, the requested service is not available or a host or router could not be reached.
- Since the IP protocol lacks an error-reporting or error-correcting mechanism, information is communicated via a message.

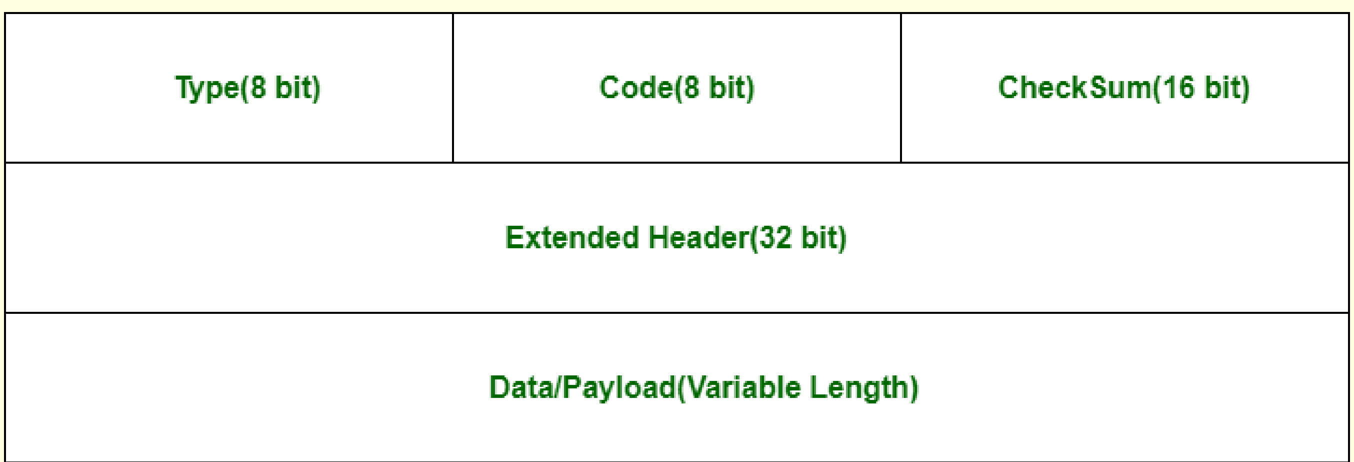
Network Layer - ICMP

Usages of ICMP:

- ICMP is used for error reporting if two devices connect over the internet and some error occurs, So, the router sends an ICMP error message to the source informing about the error.
- ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility.
  - Traceroute: Traceroute utility is used to know the route between two devices connected over the internet. It routes the journey from one router to another, and a traceroute is performed to check network issues before data transfer.
  - Ping: Ping is a simple kind of traceroute known as the echo-request message, it is used to measure the time taken by data to reach the destination and return to the source, these replies are known as echo-replies messages.

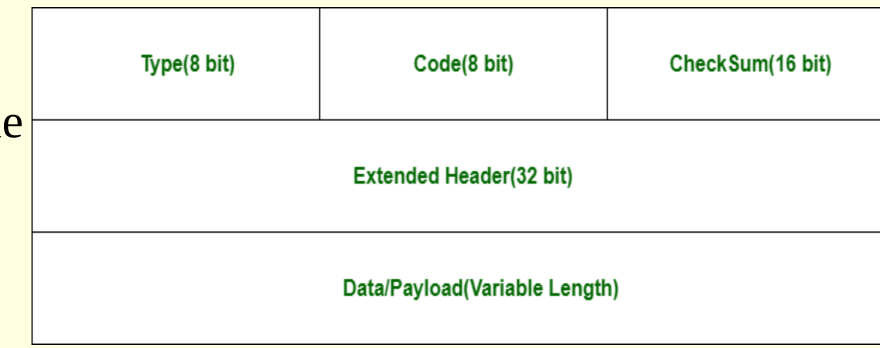
Network Layer - ICMP

- How Does ICMP Work?
- ICMP is the primary and important protocol of the IP suite, but ICMP isn’t associated with any transport layer protocol (TCP or UDP) as it doesn’t need to establish a connection with the destination device before sending any message as it is a connectionless protocol.
- The working of ICMP is just contrasting with TCP, as TCP is a connection-oriented protocol whereas ICMP is a connectionless protocol. Whenever a connection is established before the message sending, both devices must be ready through a TCP Handshake.
- ICMP packets are transmitted in the form of datagrams that contain an IP header with ICMP data. ICMP datagram is similar to a packet, which is an independent data entity.



Network Layer - ICMP

- ICMP Packet Format - ICMP header comes after IPv4 and IPv6 packet header.
- In the ICMP packet format, the first 32 bits of the packet contain three fields:
- Type (8-bit): The initial 8-bit of the packet is for message type, it provides a brief description of the message so that receiving network would know what kind of message it is receiving and how to respond to it. Some common message types are as follows:
  - Type 0 – Echo reply
  - Type 3 – Destination unreachable
  - Type 5 – Redirect Message
  - Type 8 – Echo Request
  - Type 11 – Time Exceeded
  - Type 12 – Parameter problem



- Code (8-bit): Code is the next 8 bits of the ICMP packet format, this field carries some additional information about the error message and type.
- Checksum (16-bit): Last 16 bits are for the checksum field in the ICMP packet header. The checksum is used to check the number of bits of the complete message and enable the ICMP tool to ensure that complete data is delivered.

Network Layer - ICMP

The next 32 bits of the ICMP Header are Extended Header which has the work of pointing out the problem in IP Message. Byte locations are identified by the pointer which causes the problem message and receiving device looks here for pointing to the problem.

The last part of the ICMP packet is Data or Payload of variable length. The bytes included in IPv4 are 576 bytes and in IPv6, 1280 bytes.

Network Layer - ICMP

Types of ICMP Messages

Type	Code	Description
0 – Echo Reply	0	Echo reply
3 – Destination Unreachable	0	Destination network unreachable
	1	Destination host unreachable
	2	Destination protocol unreachable
	3	Destination port unreachable
	4	Fragmentation is needed and the DF flag set
	5	Source route failed
5 – Redirect Message	0	Redirect the datagram for the network
	1	Redirect datagram for the host
	2	Redirect the datagram for the Type of Service and Network
	3	Redirect datagram for the Service and Host
8 – Echo Request	0	Echo request
9 – Router Advertisement	0	Use to discover the addresses of operational routers

Network Layer- IGMP

- IGMP is acronym for Internet Group Management Protocol. IGMP **is a communication protocol used by hosts and adjacent routers for multicasting communication with IP networks and uses the resources efficiently to transmit the message/data packets.** Multicast communication can have single or multiple senders and receivers and thus, IGMP can be used in streaming videos, gaming or web conferencing tools. This protocol is used on IPv4 networks and for using this on IPv6, multicasting is managed by Multicast Listener Discovery (MLD). Like other network protocols, IGMP is used on network layer. MLDv1 is almost same in functioning as IGMPv2 and MLDv2 is almost similar to IGMPv3. The communication protocol, IGMPv1 was developed in 1989 at Stanford University. IGMPv1 was updated to IGMPv2 in year 1997 and again updated to IGMPv3 in year 2002.

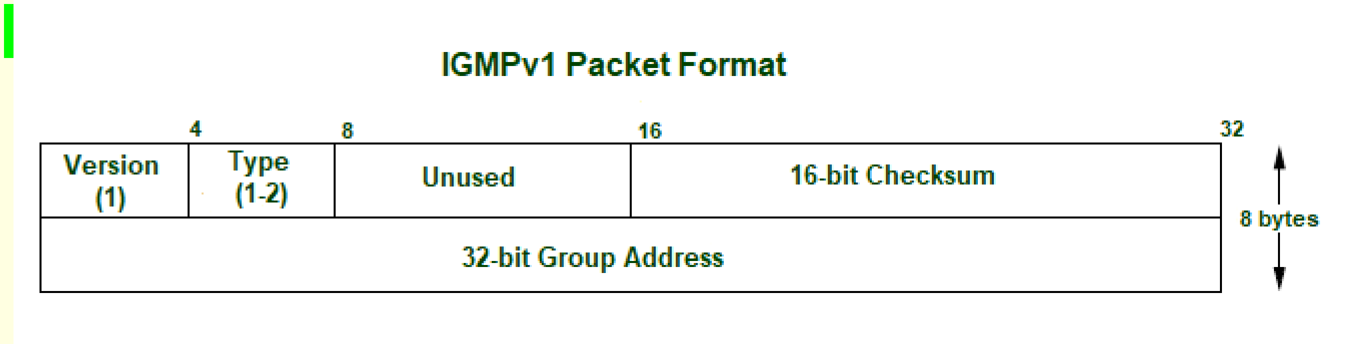
Network Layer- IGMP

- The IGMP protocol is used by the hosts and router to identify the hosts in a LAN that are the members of a group. IGMP is a part of the IP layer and IGMP has a fixed- size message. The IGMP message is encapsulated within an IP datagram.
- The IP protocol supports two types of communication:
- Unicasting: It is a communication between one sender and one receiver. Therefore, we can say that it is one-to-one communication.
- Multicasting: Sometimes the sender wants to send the same message to a large of receivers simultaneously. This process is known as multicasting which has one-to-many communication.

Network Layer- IGMP

- Types: There are 3 versions of IGMP. These versions are backward compatible. Following are the versions of IGMP:

1. IGMPv1: The version of IGMP communication protocol allows all the supporting hosts to join the multicast groups using membership request and include some basic features. But, host cannot leave the group on their own and have to wait for a timeout to leave the group. The message packet format in IGMPv1:

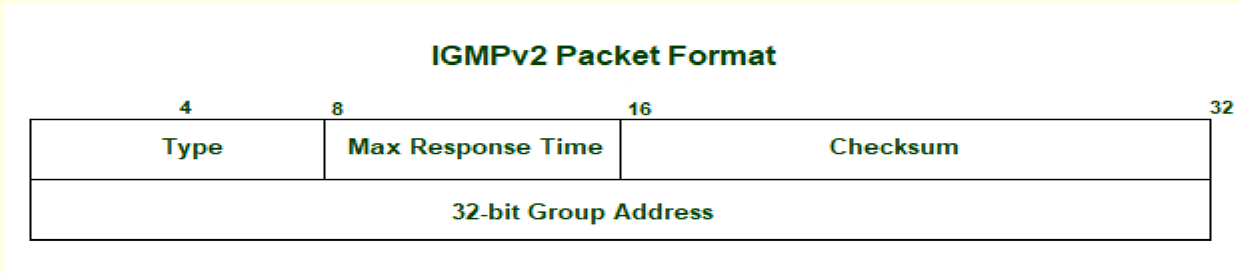


- Version – Set to 1.
- Type – 1 for Host Membership Query and Host Membership Report.
- Unused – 8-bits of zero which are of no use.
- Checksum – It is the one's complement of the sum of IGMP messages.
- Group Address – The group address field is zero when sent and ignored when received in membership query message. In a membership report message, the group address field takes the IP host group address of the group being reported.

Network Layer- IGMP

2. IGMPv2: IGMPv2 is the revised version of IGMPv1 communication protocol. It has added functionality of leaving the multicast group using group membership.

The message packet format in IGMPv2:



- Max Response Time: This field is ignored for message types other than membership query. For membership query type, it is the maximum time allowed before sending a response report. The value is in units of 0.1 seconds.
- Checksum: It is the one's complement of the sum of IGMP message. It determines the entire payload of the IP datagram in which IGMP message is encapsulated.
- Group Address: It is set as 0 when sending a general query. Otherwise, multicast address for group-specific or source-specific queries. The behavior of this field depends on the type of the message sent.

For Membership Query, the group address is set to zero for General Query and set to multicast group address for a specific query. For Membership Report, the group address is set to the multicast group address. For Leave Group, it is set to the multicast group address.

Network Layer- IGMP

3. IGMPv3: IGMPv2 was revised to IGMPv3 and added source-specific multicast and membership report aggregation. These reports are sent to 224.0.0.22. The message packet format in IGMPv3:

IGMPv3 Packet Format

Bit Offset	0-3	4	5-7	8-15	16-31
0	Type = 0x11			Max Response Code	Checksum
32	Group Address				
64	Resv	S	QRV	QQIC	Number of Sources (N)
96	Source Address[1]				
128	Source Address[2]				
	Source Address[N]				

- Max Response Time: This field is ignored for message types other than membership query. For membership query type, it is the maximum time allowed before sending a response report. The value is in units of 0.1 seconds.
- Checksum: It is the one’s complement of the one’s complement of the sum of IGMP message.
- Group Address: It is set as 0 when sending a general query. Otherwise, multicast address for group-specific or source-specific queries.
- Resv: It is set zero of sent and ignored when received.

Network Layer- IGMP

3. IGMPv3: IGMPv2 was revised to IGMPv3 and added source-specific multicast and membership report aggregation. These reports are sent to 224.0.0.22. The message packet format in IGMPv3:

IGMPv3 Packet Format

Bit Offset	0-3	4	5-7	8-15	16-31
0	Type = 0x11			Max Response Code	Checksum
32	Group Address				
64	Resv	S	QRV	QQIC	Number of Sources (N)
96	Source Address[1]				
128	Source Address[2]				
	Source Address[N]				

- S flag: It represents Suppress Router-side Processing flag. When the flag is set, it indicates to suppress the timer updates that multicast routers perform upon receiving any query.
- QRV: It represents Querier's Robustness Variable. Routers keeps on retrieving the QRV value from the most recently received query as their own value until the most recently received QRV is zero.
- QQIC: It represents Querier's Query Interval Code.
- Number of sources: It represents the number of source addresses present in the query. For general query or group-specific query, this field is zero and for group-and-source-specific query, this field is non-zero.
- Source Address[i]: It represents the IP unicast address for N fields.

Network Layer – Classes of routing protocols

- Routing is a process in which the layer 3 devices (either router or layer 3 switches) find the optimal path to deliver a packet from one network to another. Dynamic routing protocols use metric, cost, and hop count to identify the best path from the path available for the destination network. There are mainly 3 different classes of routing protocols:

1. Distance Vector Routing Protocol

- These protocols select the best path on the basis of hop counts to reach a destination network in a particular direction. Dynamic protocol like RIP is an example of a distance vector routing protocol. Hop count is each router that occurs in between the source and the destination network. The path with the least hop count will be chosen as the best path.

Features:

- Updates of the network are exchanged periodically.
- Updates (routing information) is not broadcasted but shared to neighboring nodes only.
- Full routing tables are not sent in updates but only distance vector is shared.
- Routers always trust routing information received from neighbor routers. This is also known as routing on rumors.

Network Layer – Classes of routing protocols

2. Link State Routing Protocol

- These protocols know more about Internetwork than any other distance vector routing protocol. These are also known as SPF (Shortest Path First) protocol. OSPF is an example of link-state routing protocol.

Features:

- Hello, messages, also known as keep-alive messages are used for neighbor discovery and recovery.
- Concept of triggered updates is used i.e updates are triggered only when there is a topology change.
- Only that many updates are exchanged which is requested by the neighbor router.

Link state routing protocol maintains three tables namely:

- Neighbor table: the table which contains information about the neighbors of the router only, i.e, to which adjacency has been formed.
- Topology table: This table contains information about the whole topology i.e contains both best and backup routes to a particular advertised networks.
- Routing table: This table contains all the best routes to the advertised network.

Network Layer – Classes of routing protocols

3. Advanced Distance Vector Routing Protocol

- It is also known as hybrid routing protocol which uses the concept of both distance vector and link-state routing protocol.
- Enhanced Interior Gateway Routing Protocol (EIGRP) is an example of this class of routing protocol. EIGRP acts as a link-state routing protocol as it uses the concept of Hello protocol for neighbor discovery and forming an adjacency.
- Also, partial updates are triggered when a change occurs.
- EIGRP acts as a distance-vector routing protocol as it learned routes from directly connected neighbors.

Network Layer - routing information protocol (RIP)

- Routing Information Protocol (RIP) **is a dynamic routing protocol that uses hop count as a routing metric to find the best path between the source and the destination network.** It is a distance-vector routing protocol that has an administrative distance (AD) value of 120 and works on the Network layer of the OSI model. RIP uses port number 520.
- Hop Count
- **Hop count is the number of routers occurring in between the source and destination network.** The path with the lowest hop count is considered as the best route to reach a network and therefore placed in the routing table. RIP prevents routing loops by limiting the number of hops allowed in a path from source and destination. The maximum hop count allowed for RIP is 15 and a hop count of 16 is considered as network unreachable.
- Features of RIP

1. Updates of the network are exchanged periodically.
2. Updates (routing information) are always broadcast.
3. Full routing tables are sent in updates.
4. Routers always trust routing information received from neighbor routers. This is also known as Routing on rumors.

Network Layer - routing information protocol (RIP)

- RIP versions:
- There are three versions of routing information protocol – RIP Version1, RIP Version2, and RIPng.

RIP v1	**RIP v2**	**RIPng**
Sends update as broadcast	Sends update as multicast	Sends update as multicast
Broadcast at 255.255.255.255	Multicast at 224.0.0.9	Multicast at FF02::9 (RIPng can only run on IPv6 networks)
Doesn't support authentication of updated messages	Supports authentication of RIPv2 update messages	–
Classful routing protocol	Classless protocol updated supports classful	Classless updates are sent

- RIP v1 is known as Classful Routing Protocol because it doesn't send information of subnet mask in its routing update.
- RIP v2 is known as Classless Routing Protocol because it sends information of subnet mask in its routing update.

Network Layer - routing information protocol (RIP)

- >> Use debug command to get the details:

  # debug ip rip

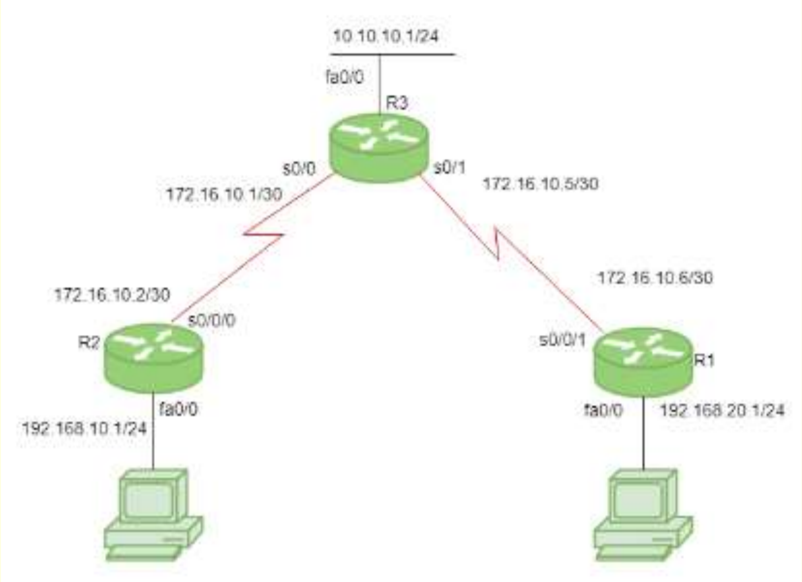
- >> Use this command to show all routes configured in router, say for router R1:

  R1# show ip route

- >> Use this command to show all protocols configured in router, say for router R1:
-
  R1# show ip protocols

Network Layer - routing information protocol (RIP)

- Configuration:
- Consider the above-given topology which has 3-routers R1, R2, R3. R1 has IP address 172.16.10.6/30 on s0/0/1, 192.168.20.1/24 on fa0/0. R2 has IP address 172.16.10.2/30 on s0/0/0, 192.168.10.1/24 on fa0/0. R3 has IP address 172.16.10.5/30 on s0/1, 172.16.10.1/30 on s0/0, 10.10.10.1/24 on fa0/0.



- Configure RIP for R1:

```
R1(config)# router rip
R1(config-router)# network 192.168.20.0
R1(config-router)# network 172.16.10.4
R1(config-router)# version 2
R1(config-router)# no auto-summary
```

- Note: no auto-summary command disables the auto-summarisation. If we don’t select any auto-summary, then the subnet mask will be considered as classful in Version 1.

Network Layer - routing information protocol (RIP)

- Configuring RIP for R2:

```
R2(config)# router rip
R2(config-router)# network 192.168.10.0
R2(config-router)# network 172.16.10.0
R2(config-router)# version 2
R2(config-router)# no auto-summary
```

- Similarly, Configure RIP for R3:

```
R3(config)# router rip
R3(config-router)# network 10.10.10.0
R3(config-router)# network 172.16.10.4
R3(config-router)# network 172.16.10.0
R3(config-router)# version 2
R3(config-router)# no auto-summary
```

Network Layer - routing information protocol (RIP)

- RIP timers:
- Update timer: The default timing for routing information being exchanged by the routers operating RIP is 30 seconds. Using an Update timer, the routers exchange their routing table periodically.
- Invalid timer: If no update comes until 180 seconds, then the destination router considers it invalid. In this scenario, the destination router mark hop counts as 16 for that router.
- Hold down timer: This is the time for which the router waits for a neighbor router to respond. If the router isn't able to respond within a given time then it is declared dead. It is 180 seconds by default.
- Flush time: It is the time after which the entry of the route will be flushed if it doesn't respond within the flush time. It is 60 seconds by default. This timer starts after the route has been declared invalid and after 60 seconds i.e time will be 180 + 60 = 240 seconds.

- Note that all these times are adjustable. Use this command to change the timers:

```
R1(config-router)#  timers basic
R1(config-router)#  timers basic 20   80    80    90
```

Network Layer - open shortest path first (OSPF)

- Open Shortest Path First (OSPF) **is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First.**
- Router Id: It is the highest active IP address present on the router. First, the highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.
- Router priority: It is an 8-bit value assigned to a router operating OSPF, used to elect DR and BDR in a broadcast network.
- Designated Router (DR): It is elected to minimize the number of adjacencies formed. DR distributes the LSAs to all the other routers. DR is elected in a broadcast network to which all the other routers share their DBD. In a broadcast network, the router requests for an update to DR, and DR will respond to that request with an update.

Network Layer - open shortest path first (OSPF)

- Open Shortest Path First (OSPF) **is a link-state routing protocol that is used to find the best path between the source and the destination router using its own Shortest Path First.**

- Backup Designated Router (BDR): BDR is a backup to DR in a broadcast network. When DR goes down, BDR becomes DR and performs its functions.

- DR and BDR election: DR and BDR election takes place in the broadcast network or multi-access network. Here are the criteria for the election:
  - The router having the highest router priority will be declared as DR.
  - If there is a tie in router priority then the highest router Id be considered. First, the highest loopback address is considered. If no loopback is configured then the highest active IP address on the interface of the router is considered.

Network Layer - open shortest path first (OSPF)

- OSPF States - The device operating OSPF goes through certain states. These states are:
- Down: In this state, no hello packets have been received on the interface.
  - Note – The Downstate doesn't mean that the interface is physically down. Here, it means that the OSPF adjacency process has not started yet.
- INIT: In this state, the hello packets have been received from the other router.
- 2WAY: In the 2WAY state, both the routers have received the hello packets from other routers. Bidirectional connectivity has been established.
  - Note – In between the 2WAY state and Exstart state, the DR and BDR election takes place.
- Exstart: In this state, NULL DBD (database description) are exchanged. In this state, the master and slave elections take place. The router having the higher router Id become the master while the other becomes the slave. This election decides Which router will send its DBD first (routers who have formed neighbourship will take part in this election).

Network Layer - open shortest path first (OSPF)

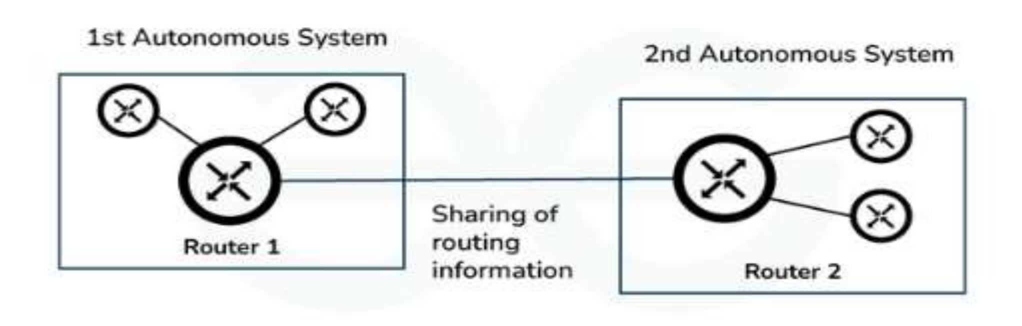
- OSPF States…
- Exchange: In this state, the actual DBDs (database descriptions) are exchanged.
- Loading: In this state, LSR (Link State Request), LSU (Link State Update), and LSA (Link State Acknowledgement) are exchanged.
- Important: When a router receives DBD from other router, it compares its own DBD with the other router DBD. If the received DBD is more updated than its own DBD then the router will send LSR to the other router stating what links are needed. The other router replies with the LSU containing the updates that are needed. In return to this, the router replies with the LSA (Link State Acknowledgement).
- Full: In this state, synchronization of all the information takes place. OSPF routing can begin only after the Full state.

Network Layer - border gateway protocol (BGP)

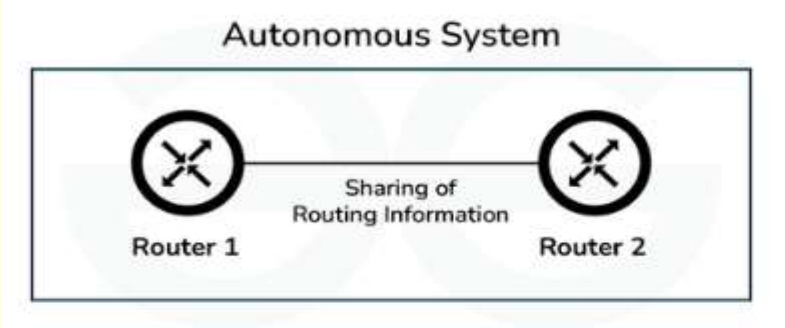
- The protocol can connect any internetwork of the autonomous system (AS) using an arbitrary topology. The only requirement is that each AS have at least one router that can run BGP and that is the router connected to at least one other AS's BGP router. BGP's main function is to exchange network reachability information with other BGP systems. Border Gateway Protocol **constructs an autonomous systems graph based on the information exchanged between BGP routers.**
- Characteristics of Border Gateway Protocol (BGP)
- Inter-Autonomous System Configuration: The main role of BGP is to provide communication between two autonomous systems.
- BGP supports the Next-Hop Paradigm.
- Coordination among multiple BGP speakers within the AS (Autonomous System).
- Path Information: BGP advertisements also include path information, along with the reachable destination and next destination pair.
- Policy Support: BGP can implement policies that can be configured by the administrator. For ex:- a router running BGP can be configured to distinguish between the routes that are known within the AS and that which are known from outside the AS.

Network Layer - border gateway protocol (BGP)

- Types of Border Gateway Protocol
- External BGP: It is used to interchange routing information between the routers in different autonomous systems, it is also known as eBGP(External Border Gateway Protocol). The below image shows how eBGP interchange routing information.



- Internal BGP: It is used to interchange routing information between the routers in the same autonomous system, it is also known as iBGP(Internal Border Gateway Protocol). Internal routers also ensure consistency among routers for sharing routing information. The below image shows how iBGP interchange routing information.



Network Layer - border gateway protocol

- Elements of BGP
- Some elements of to each path and these elements help routers to select a path from multiple paths. Here below are some elements of BGP:
- Weight: Weight is defined as a Cisco-specific attribute that tells a router which path is preferred. The weight having a higher value is preferred.
- Originate: This tells how a router choose routes and adds to BGP itself.
- Local Preference: Local Preference is an element used to select the outbound routing path. Greater local preference is preferred.
- Autonomous System Path: This element tells the router to select a path having a shorter length.
- Next Hop: To reach the destination the next hop elements specify the IP address that should be used as the next hop.
- BGP Route Information Management Functions:
- Route Storage: Each BGP stores information about how to reach other networks.
- Route Update: In this task, Special techniques are used to determine when and how to use the information received from peers to properly update the routes.
- Route Selection: Each BGP uses the information in its route databases to select good routes to each network on the internet network.
- Route advertisement: Each BGP speaker regularly tells their peer what is known about various networks and methods to reach them.

Network Layer - border gateway protocol (BGP)

- Difference Between BGP and OSPF

BGP	**OSPF**
It follows the Path Vector Routing Algorithm	It follows the Link State Routing Algorithm
The speed of convergence is very slow in BGP	The speed of convergence is fast in the case of OSPF
BGP is also called inter-domain routing protocol	OSPF is also called intra-domain routing protocol
In BGP routing operation is performed between two autonomous system (AS)	In OSPF routing operation is performed inside an autonomous system (AS)
In BGP, TCP protocol is used	In OSPF, IP protocol is used

Network Layer - Enhanced Interior Gateway Routing Protocol (EIGRP)

- Enhanced Interior Gateway Routing Protocol (EIGRP) **is a dynamic routing protocol that is used to find the best path between any two-layer 3 devices to deliver the packet**. EIGRP works on network layer Protocol of OSI model and uses protocol number 88. It uses metrics to find out the best path between two layer 3 devices (router or layer 3 switches) operating EIGRP. Administrative Distance for EIGRP are:-

EIGRP routes	AD values
Summary Routes	5
Internal Routes	90
external routes	170

It uses some messages to communicate with the neighbour devices that operate EIGRP. These are:-

- Hello message: These messages are kept alive messages which are exchanged between two devices operating EIGRP. These messages are used for neighbour discovery/recovery, if there is any device operating EIGRP or if any device(operating EIGRP) coming up again.

These messages are used for neighbor discovery if multicast at 224.0.0.10. It contains values like AS number, k values, etc.

These messages are used as acknowledgement when unicast. A hello with no data is used as the acknowledgement.

Network Layer – EIGRP Messages

- NULL update: It is used to calculate SRTT (Smooth Round Trip Timer) and RTO(Retransmission Time Out).
  - SRTT: The time is taken by a packet to reach the neighboring router and the acknowledgement of the packet to reach the local router.
  - RTO: If a multicast fails then unicast is being sent to that router. RTO is the time for which the local router waits for an acknowledgement of the packet.
- Full Update: After exchanging hello messages or after the neighbourship is formed, these messages are exchanged. This message contains all the best routes.
- Partial update: These messages are exchanged when there is a topology change and new links are added. It contains only the new routes, not all the routes. These messages are multicast.

Network Layer – EIGRP Messages

- Query message: These messages are multicast when the device is declared dead and it has no routes to it in its topology table.
- Reply message: These messages are the acknowledgment of the query message sent to the originator of the query message stating the route to the network which has been asked in the query message.
- Acknowledgement message:

It is used to acknowledge EIGRP updates, queries, and replies. Acks are hello packets that contain no data.

Note:- Hello and acknowledgment packets do not require any acknowledgment.

Reply, query, update messages are reliable messages i.e require acknowledgement.

Network Layer - Enhanced Interior Gateway Routing Protocol (EIGRP)

- Composite matrix: The EIGRP composite metric calculation can use up to 5 variables, but only 2 are used by default (K1 and K3).
- The composite metric values are:
  - K1 (bandwidth)
  - K2 (load)
  - K3 (delay)
  - K4 (reliability)
  - K5 (MTU)
- The lowest bandwidth, load, delay, reliability, MTU along the path between the source and the destination is considered in the composite matrix in order to calculate the cost.

- Note:- Generally, only k1 and k3 values are used for metric calculation by EIGRP. The values are 10100 for k1, k2, k3, k4, k5 respectively.

Network Layer - Enhanced Interior Gateway Routing Protocol (EIGRP)

- Criteria to form EIGRP neighbourship, these criteria should be fulfilled:-
  - k values should match.
  - Autonomous system number should match. (AS is a group of networks running under a single administrative control) .
  - authentication should match (if applied). EIGRP supports MD5 authentication only.
  - subnet mask should be the same.

- EIGRP Timers:-
  - Hello timer: The interval in which EIGRP sends a hello message on an interface. It is 5 seconds by default.
  - Dead timer: The interval in which the neighbor will be declared dead if it is not able to send the hello packet. It is 15 seconds by default.

Computer Networks and Data Communication

2026